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(54) **INDICATIONS AND REPORTING FOR
BACKSCATTER READING OPERATIONS**

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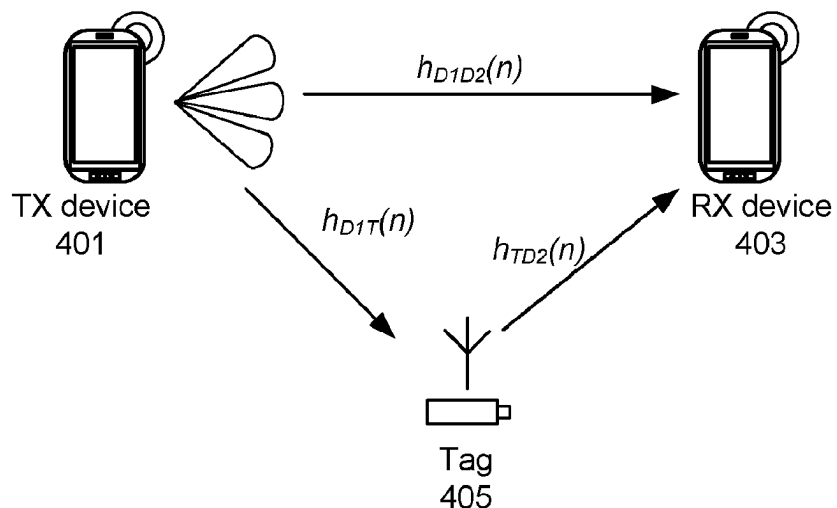
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(57)

ABSTRACT

Various aspects of the present disclosure generally relate to wireless communication. In some aspects, a receiving device may receive an indication of an upcoming backscatter reading operation. Accordingly, the receiving device may adjust a reception parameter of the receiving device in response to the indication. Numerous other aspects are described.

400 →



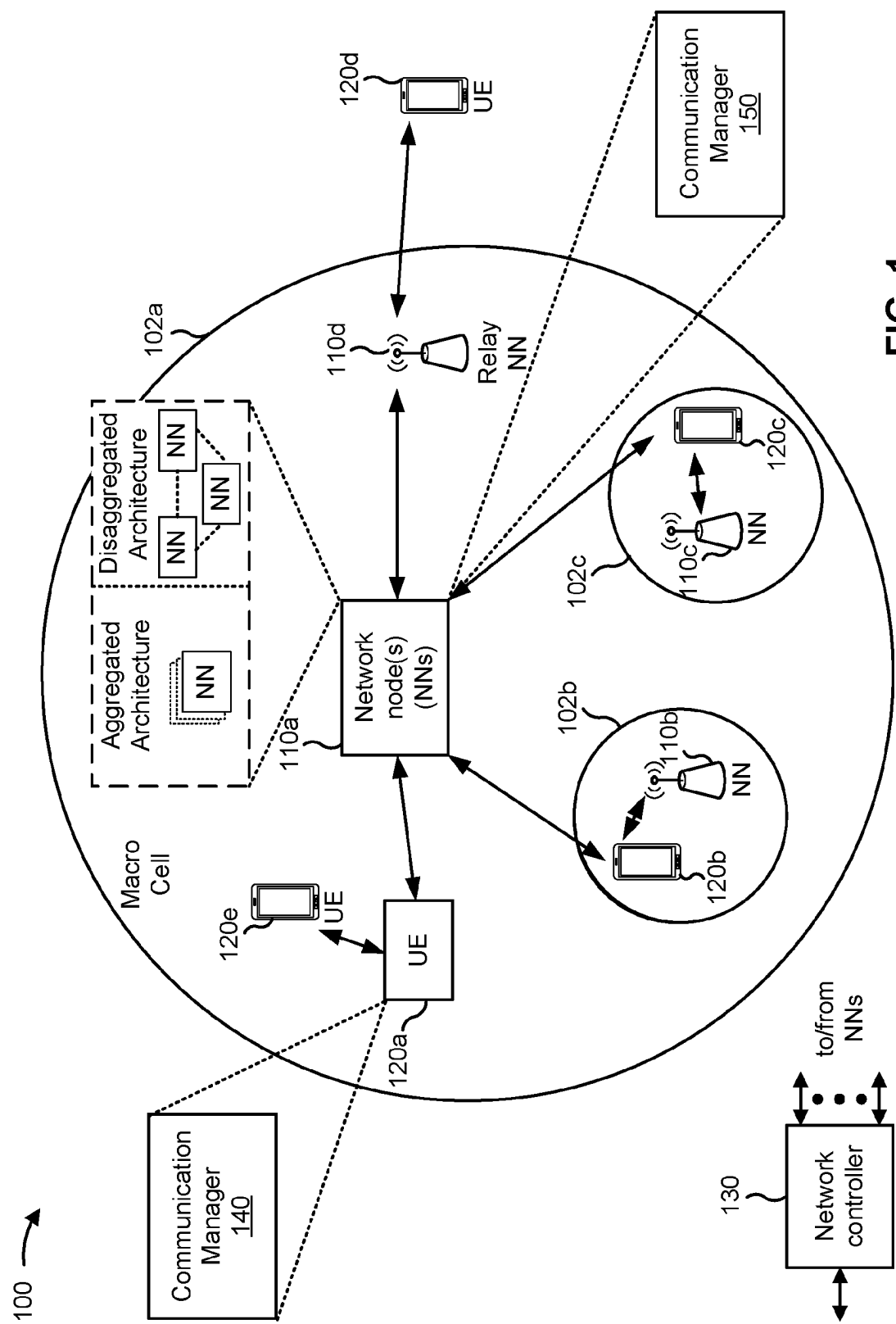


FIG. 1

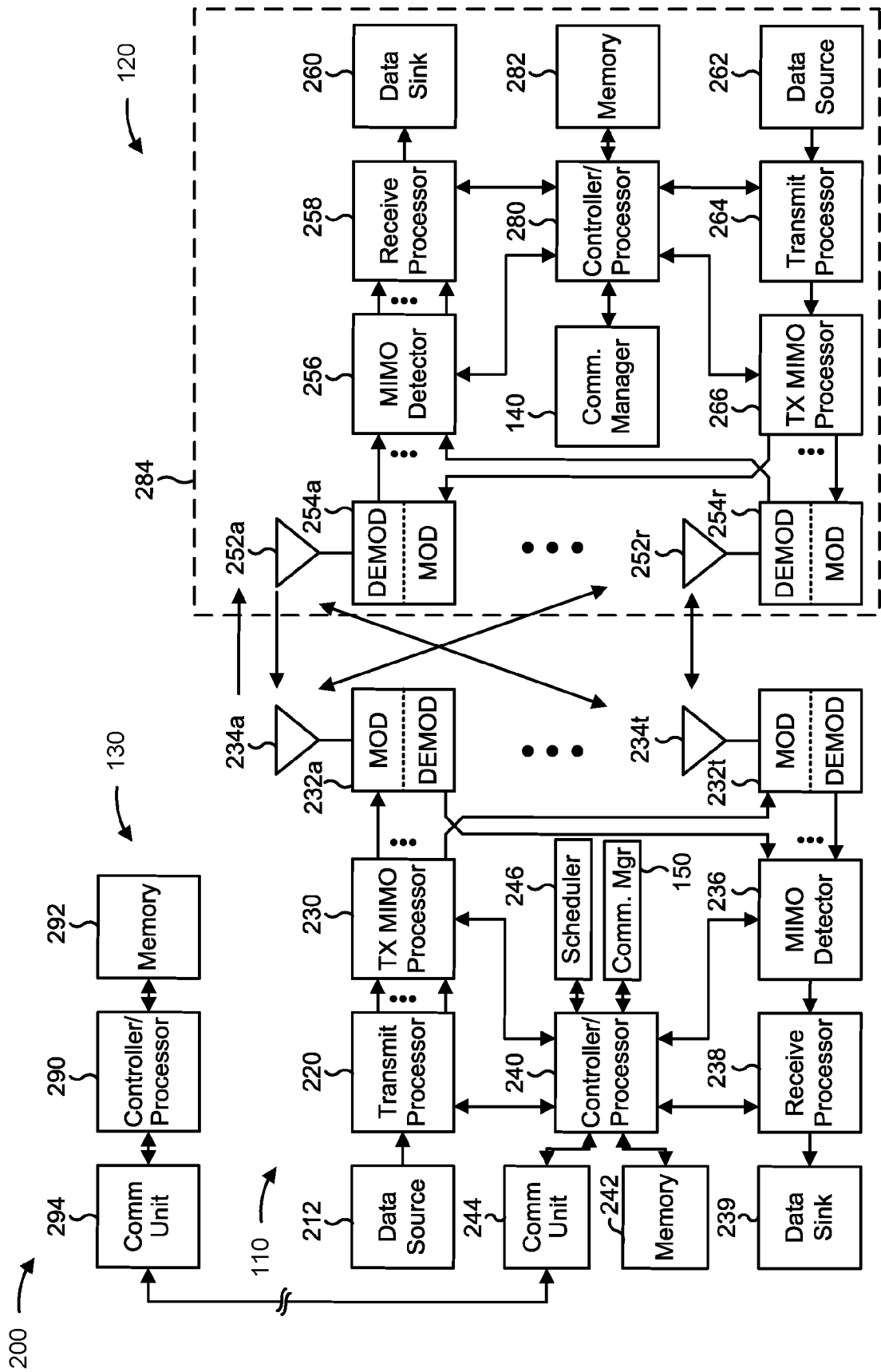


FIG. 2

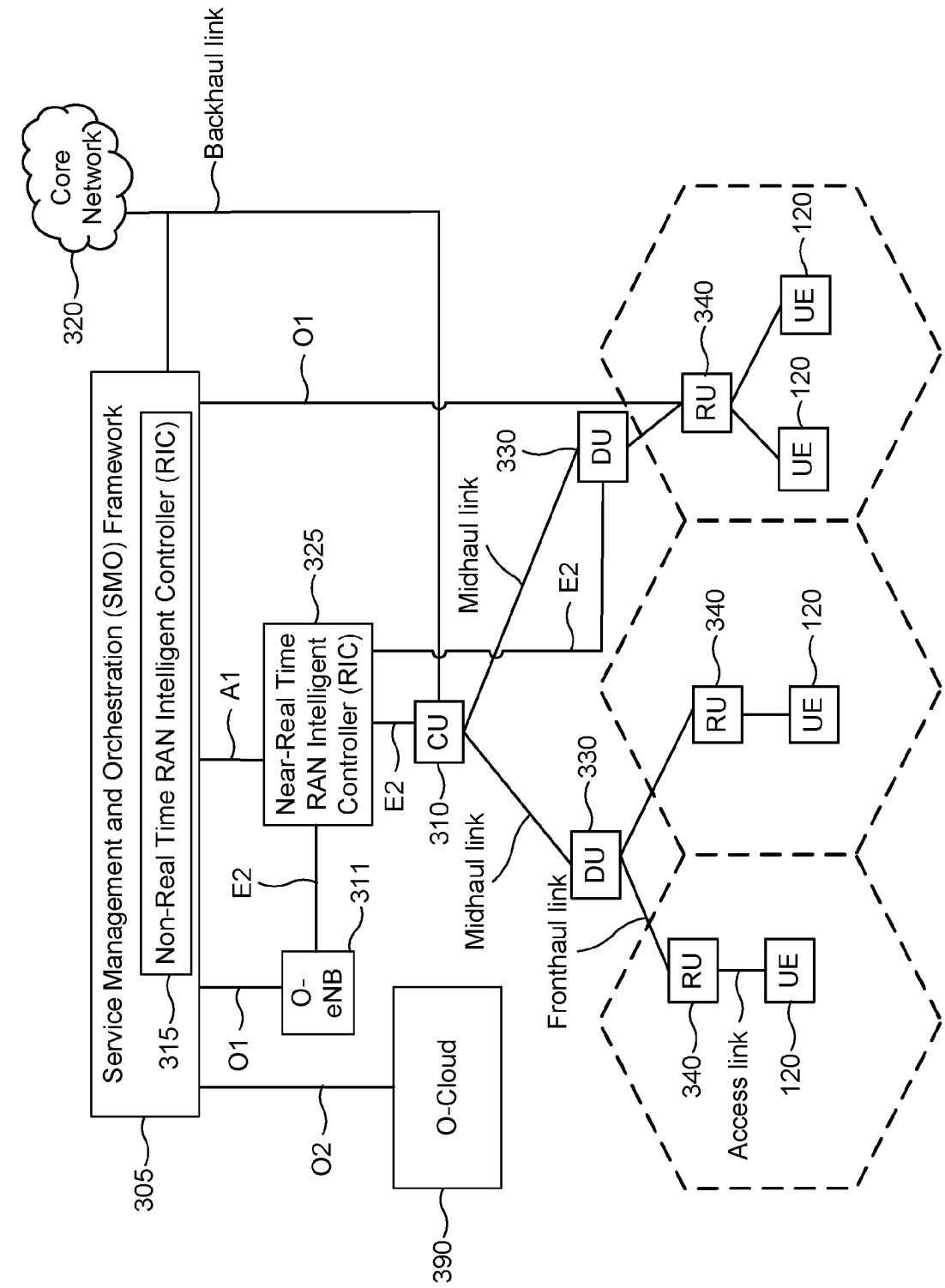


FIG. 3

400

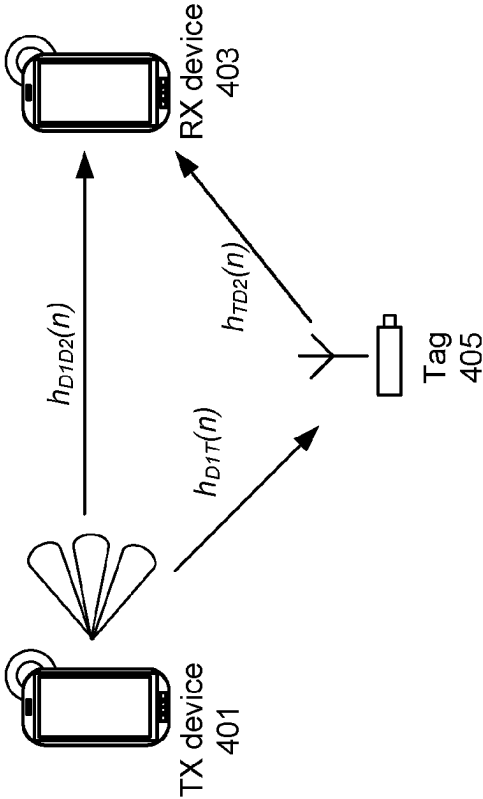


FIG. 4

500

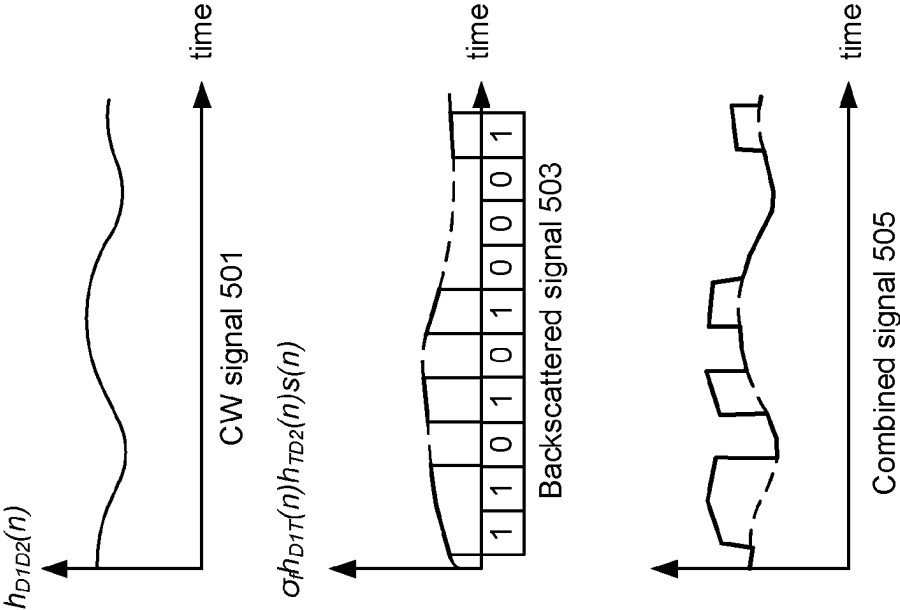


FIG. 5

600

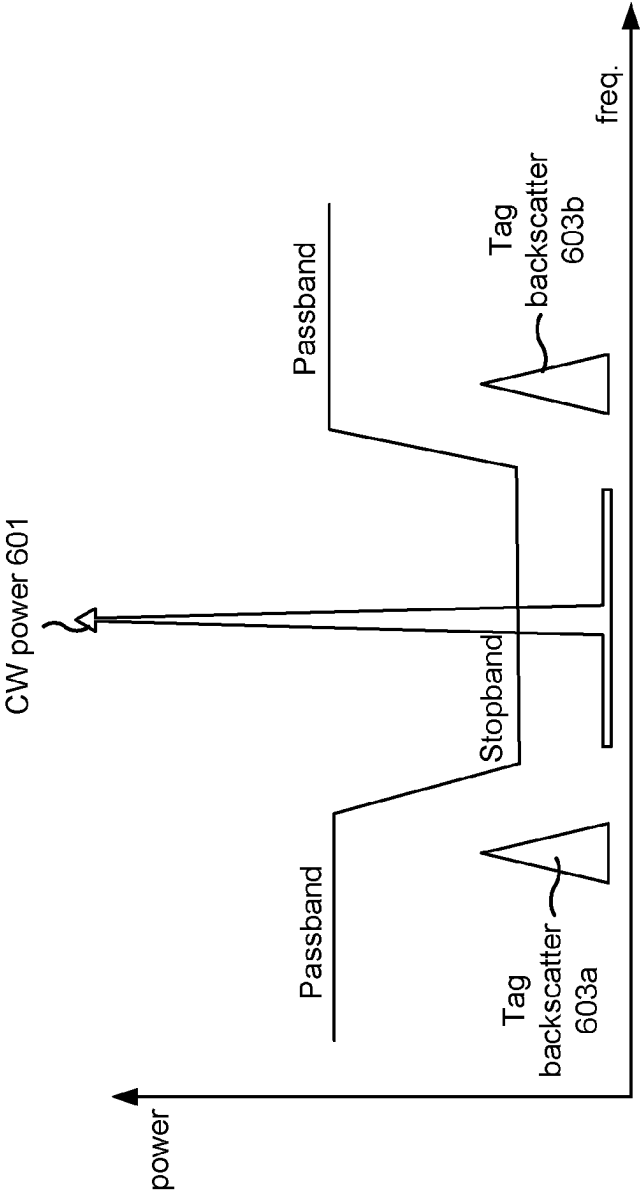


FIG. 6

700

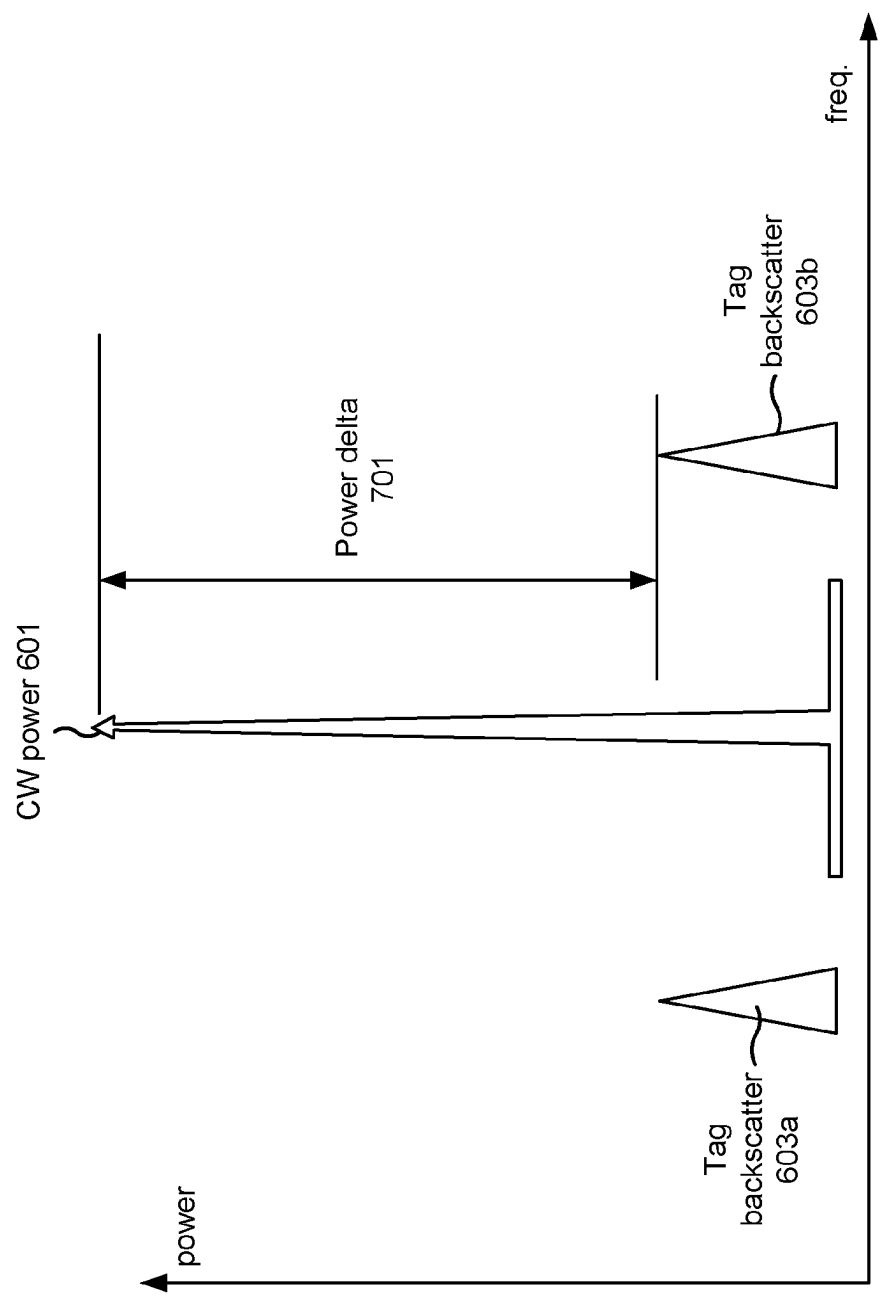


FIG. 7

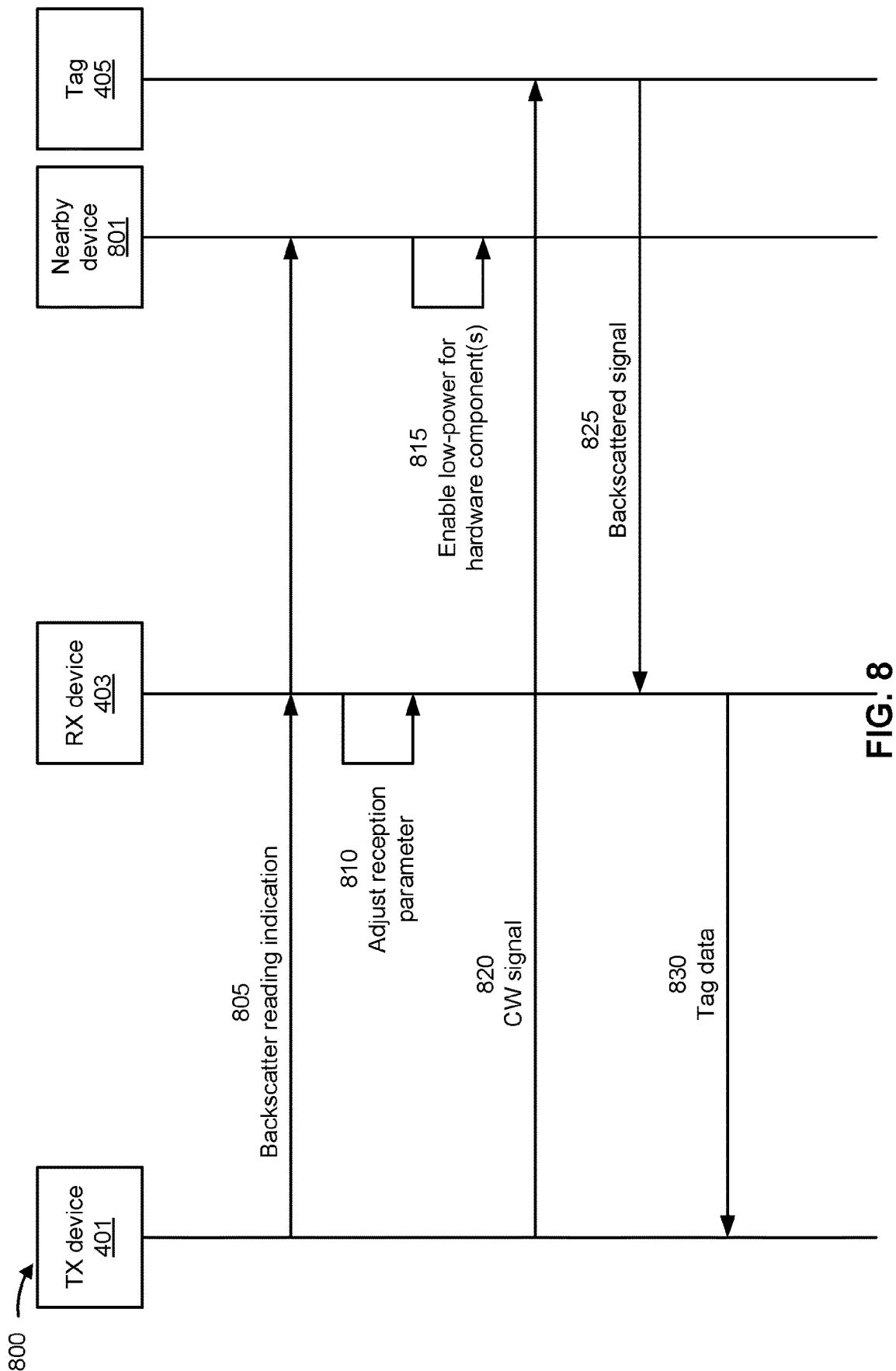


FIG. 8

900 ↗

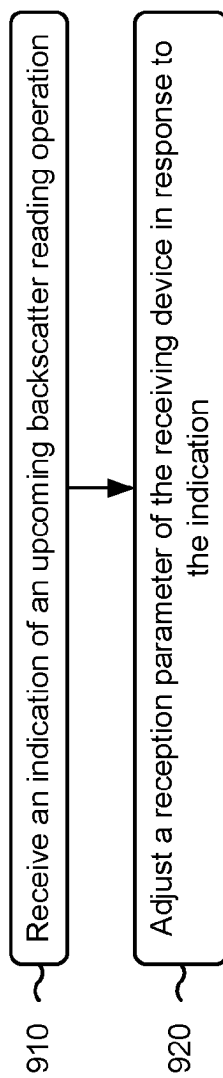


FIG. 9

1000 →

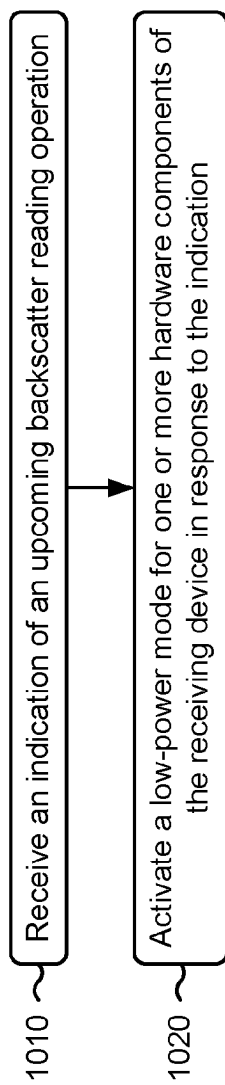


FIG. 10

1100 →

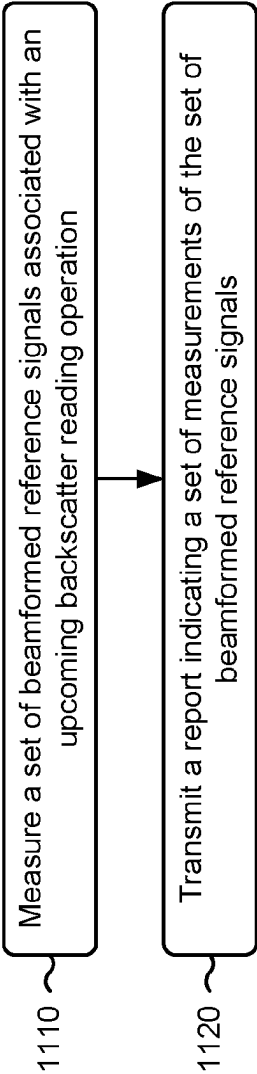


FIG. 11

1200 ↗

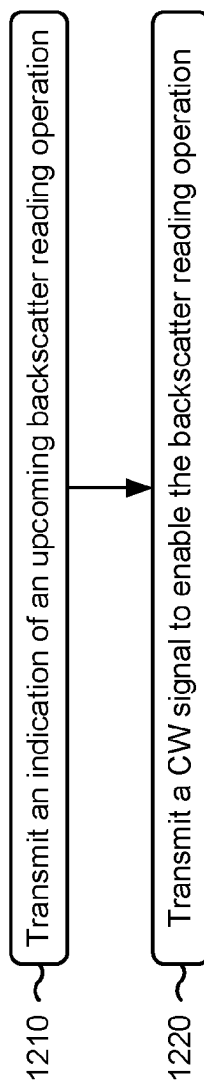


FIG. 12

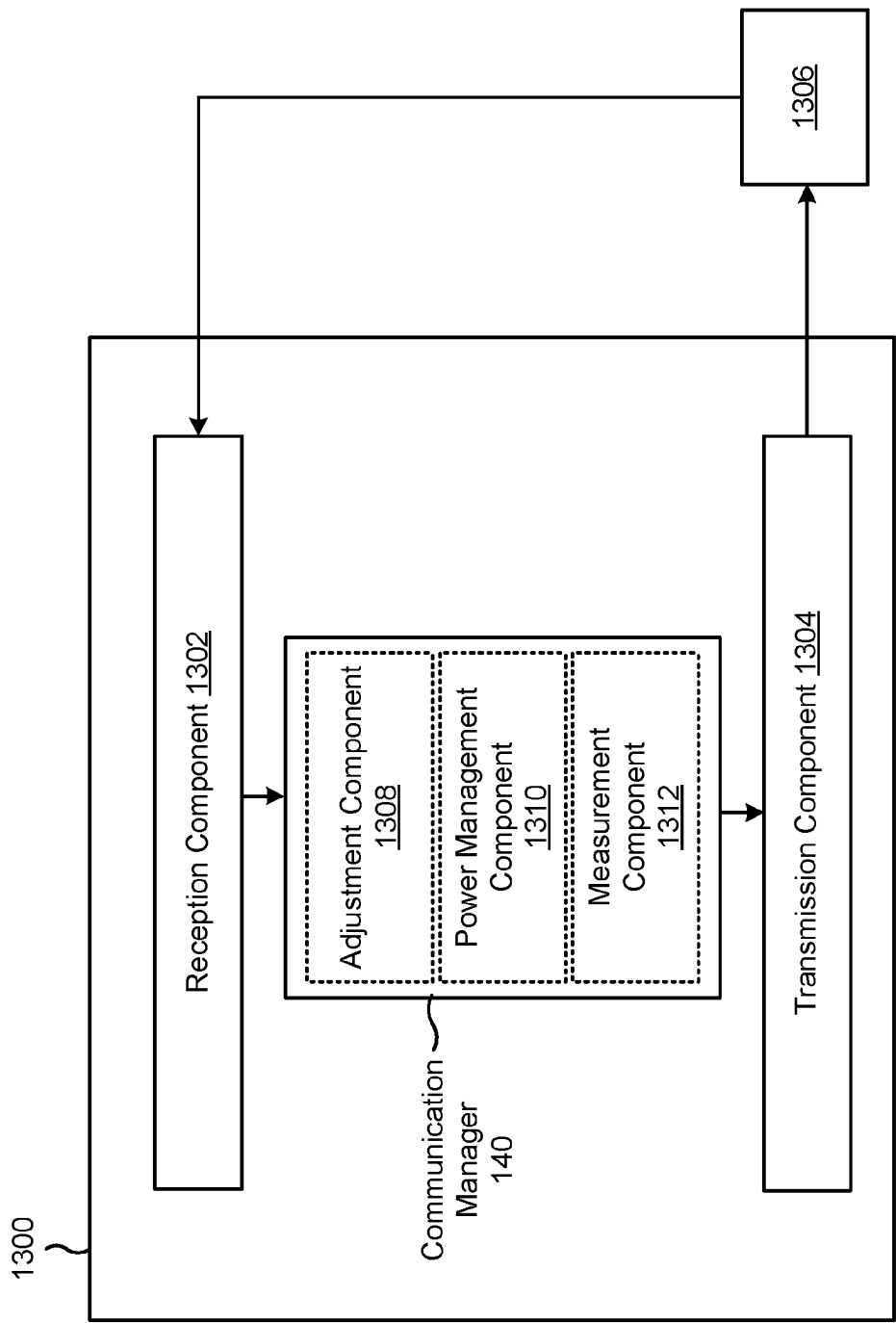


FIG. 13

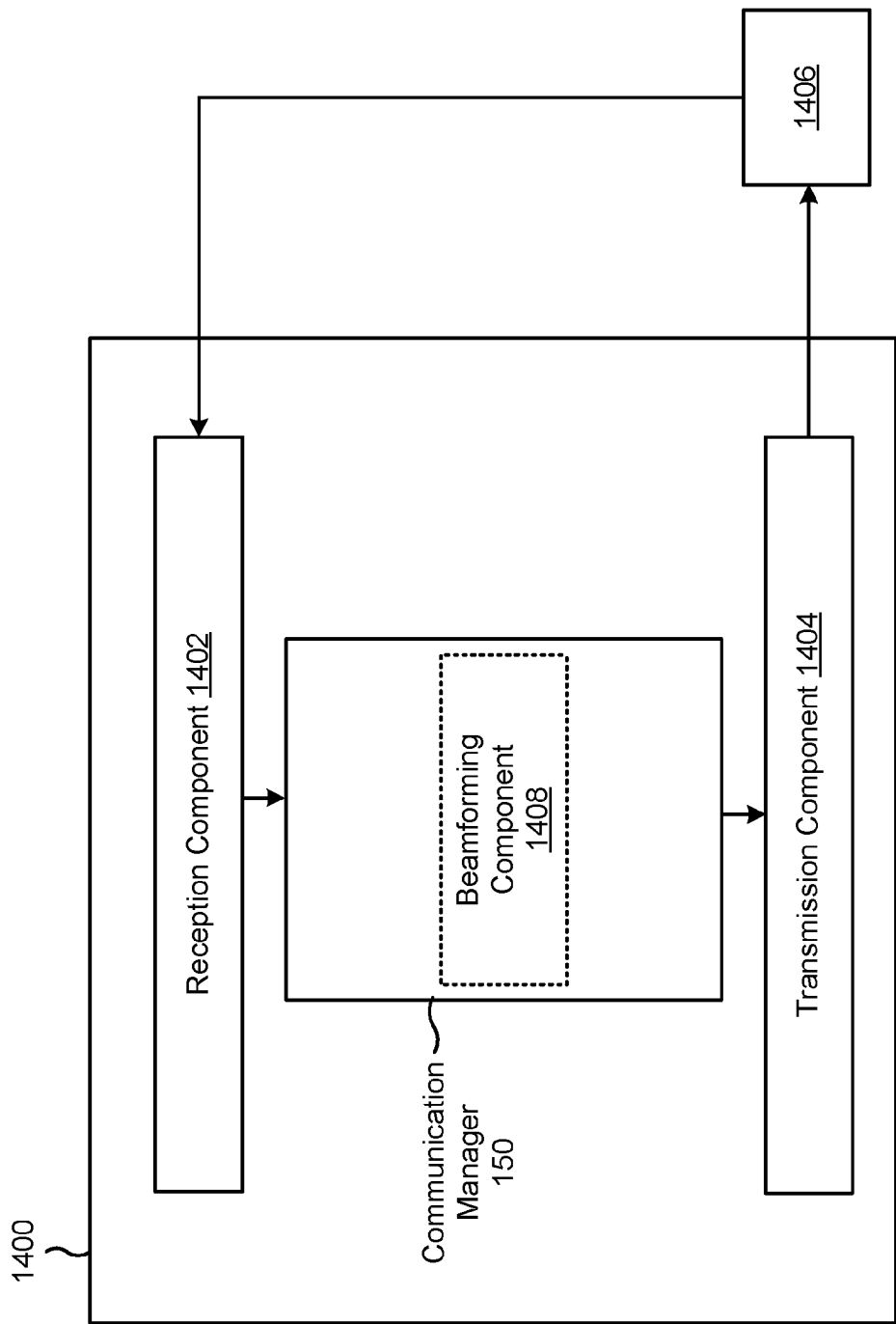


FIG. 14

INDICATIONS AND REPORTING FOR BACKSCATTER READING OPERATIONS

FIELD OF THE DISCLOSURE

[0001] Aspects of the present disclosure generally relate to wireless communication and to techniques and apparatuses for indications and reporting for backscatter reading operations.

BACKGROUND

[0002] Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources (e.g., bandwidth, transmit power, or the like). Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, time division synchronous code division multiple access (TD-SCDMA) systems, and Long Term Evolution (LTE). LTE/LTE-Advanced is a set of enhancements to the Universal Mobile Telecommunications System (UMTS) mobile standard promulgated by the Third Generation Partnership Project (3GPP).

[0003] A wireless network may include one or more network nodes that support communication for wireless communication devices, such as a user equipment (UE) or multiple UEs. A UE may communicate with a network node via downlink communications and uplink communications. “Downlink” (or “DL”) refers to a communication link from the network node to the UE, and “uplink” (or “UL”) refers to a communication link from the UE to the network node. Some wireless networks may support device-to-device communication, such as via a local link (e.g., a sidelink (SL), a wireless local area network (WLAN) link, and/or a wireless personal area network (WPAN) link, among other examples).

[0004] The above multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different UEs to communicate on a municipal, national, regional, and/or global level. New Radio (NR), which may be referred to as 5G, is a set of enhancements to the LTE mobile standard promulgated by the 3GPP. NR is designed to better support mobile broadband internet access by improving spectral efficiency, lowering costs, improving services, making use of new spectrum, and better integrating with other open standards using orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) (CP-OFDM) on the downlink, using CP-OFDM and/or single-carrier frequency division multiplexing (SC-FDM) (also known as discrete Fourier transform spread OFDM (DFT-s-OFDM)) on the uplink, as well as supporting beamforming, multiple-input multiple-output (MIMO) antenna technology, and carrier aggregation. As the demand for mobile broadband access continues to increase, further improvements in LTE, NR, and other radio access technologies remain useful.

SUMMARY

[0005] Some aspects described herein relate to a method of wireless communication performed by a receiving device. The method may include receiving an indication of an upcoming backscatter reading operation. The method may include adjusting a reception parameter of the receiving device in response to the indication.

[0006] Some aspects described herein relate to a method of wireless communication performed by a receiving device. The method may include receiving an indication of an upcoming backscatter reading operation. The method may include activating a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0007] Some aspects described herein relate to a method of wireless communication performed by a receiving device. The method may include measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation. The method may include transmitting a report indicating a set of measurements of the set of beamformed reference signals.

[0008] Some aspects described herein relate to a method of wireless communication performed by a transmitting device. The method may include transmitting an indication of an upcoming backscatter reading operation. The method may include transmitting a continuous wave (CW) signal to enable the backscatter reading operation.

[0009] Some aspects described herein relate to an apparatus for wireless communication at a receiving device. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to receive an indication of an upcoming backscatter reading operation. The one or more processors may be configured to adjust a reception parameter of the receiving device in response to the indication.

[0010] Some aspects described herein relate to an apparatus for wireless communication at a receiving device. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to receive an indication of an upcoming backscatter reading operation. The one or more processors may be configured to activate a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0011] Some aspects described herein relate to an apparatus for wireless communication at a receiving device. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to measure a set of beamformed reference signals associated with an upcoming backscatter reading operation. The one or more processors may be configured to transmit a report indicating a set of measurements of the set of beamformed reference signals.

[0012] Some aspects described herein relate to an apparatus for wireless communication at a transmitting device. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to transmit an indication of an upcoming backscatter reading operation. The one or more processors may be configured to transmit a CW signal to enable the backscatter reading operation.

[0013] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a receiving

device. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to receive an indication of an upcoming backscatter reading operation. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to adjust a reception parameter of the receiving device in response to the indication.

[0014] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a receiving device. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to receive an indication of an upcoming backscatter reading operation. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to activate a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0015] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a receiving device. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to measure a set of beamformed reference signals associated with an upcoming backscatter reading operation. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to transmit a report indicating a set of measurements of the set of beamformed reference signals.

[0016] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a transmitting device. The set of instructions, when executed by one or more processors of the transmitting device, may cause the transmitting device to transmit an indication of an upcoming backscatter reading operation. The set of instructions, when executed by one or more processors of the transmitting device, may cause the transmitting device to transmit a CW signal to enable the backscatter reading operation.

[0017] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving an indication of an upcoming backscatter reading operation. The apparatus may include means for adjusting a reception parameter of the receiving device in response to the indication.

[0018] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving an indication of an upcoming backscatter reading operation. The apparatus may include means for activating a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0019] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation. The apparatus may include means for transmitting a report indicating a set of measurements of the set of beamformed reference signals.

[0020] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for transmitting an indication of an upcoming

backscatter reading operation. The apparatus may include means for transmitting a CW signal to enable the backscatter reading operation.

[0021] Aspects generally include a method, apparatus, system, computer program product, non-transitory computer-readable medium, user equipment, base station, network entity, network node, wireless communication device, and/or processing system as substantially described herein with reference to and as illustrated by the drawings and specification.

[0022] The foregoing has outlined rather broadly the features and technical advantages of examples according to the disclosure in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter. The conception and specific examples disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages, will be better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purposes of illustration and description, and not as a definition of the limits of the claims.

[0023] While aspects are described in the present disclosure by illustration to some examples, those skilled in the art will understand that such aspects may be implemented in many different arrangements and scenarios. Techniques described herein may be implemented using different platform types, devices, systems, shapes, sizes, and/or packaging arrangements. For example, some aspects may be implemented via integrated chip embodiments or other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, and/or artificial intelligence devices). Aspects may be implemented in chip-level components, modular components, non-modular components, non-chip-level components, device-level components, and/or system-level components. Devices incorporating described aspects and features may include additional components and features for implementation and practice of claimed and described aspects. For example, transmission and reception of wireless signals may include one or more components for analog and digital purposes (e.g., hardware components including antennas, radio frequency (RF) chains, power amplifiers, modulators, buffers, processors, interleavers, adders, and/or summers). It is intended that aspects described herein may be practiced in a wide variety of devices, components, systems, distributed arrangements, and/or end-user devices of varying size, shape, and constitution.

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] So that the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be had by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain typical aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other

equally effective aspects. The same reference numbers in different drawings may identify the same or similar elements.

[0025] FIG. 1 is a diagram illustrating an example of a wireless network, in accordance with the present disclosure.

[0026] FIG. 2 is a diagram illustrating an example of a network node in communication with a user equipment (UE) in a wireless network, in accordance with the present disclosure.

[0027] FIG. 3 is a diagram illustrating an example disaggregated base station architecture, in accordance with the present disclosure.

[0028] FIGS. 4 and 5 are diagrams illustrating an example of a backscatter reading operation, in accordance with the present disclosure.

[0029] FIG. 6 is a diagram illustrating an example associated with filters for backscatter reading operations, in accordance with the present disclosure.

[0030] FIG. 7 is a diagram illustrating an example associated with reporting for backscatter reading operations, in accordance with the present disclosure.

[0031] FIG. 8 is a diagram illustrating an example associated with performing a backscatter reading operation, in accordance with the present disclosure.

[0032] FIGS. 9, 10, 11, and 12 are diagrams illustrating example processes associated with indications and reporting for backscatter reading operations, in accordance with the present disclosure.

[0033] FIGS. 13 and 14 are diagrams of example apparatuses for wireless communication, in accordance with the present disclosure.

DETAILED DESCRIPTION

[0034] Various aspects of the disclosure are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. One skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

[0035] Several aspects of telecommunication systems will now be presented with reference to various apparatuses and techniques. These apparatuses and techniques will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, modules, components, circuits, steps, processes, algorithms, or the like (collectively referred to as “elements”). These elements may be implemented using hardware, software, or combinations thereof. Whether such elements are imple-

mented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

[0036] While aspects may be described herein using terminology commonly associated with a 5G or New Radio (NR) radio access technology (RAT), aspects of the present disclosure can be applied to other RATs, such as a 3G RAT, a 4G RAT, and/or a RAT subsequent to 5G (e.g., 6G).

[0037] FIG. 1 is a diagram illustrating an example of a wireless network 100, in accordance with the present disclosure. The wireless network 100 may be or may include elements of a 5G (e.g., NR) network and/or a 4G (e.g., Long Term Evolution (LTE)) network, among other examples. The wireless network 100 may include one or more network nodes 110 (shown as a network node 110a, a network node 110b, a network node 110c, and a network node 110d), a user equipment (UE) 120 or multiple UEs 120 (shown as a UE 120a, a UE 120b, a UE 120c, a UE 120d, and a UE 120e), and/or other entities. A network node 110 is a network node that communicates with UEs 120. As shown, a network node 110 may include one or more network nodes. For example, a network node 110 may be an aggregated network node, meaning that the aggregated network node is configured to utilize a radio protocol stack that is physically or logically integrated within a single radio access network (RAN) node (e.g., within a single device or unit). As another example, a network node 110 may be a disaggregated network node (sometimes referred to as a disaggregated base station), meaning that the network node 110 is configured to utilize a protocol stack that is physically or logically distributed among two or more nodes (such as one or more central units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)).

[0038] In some examples, a network node 110 is or includes a network node that communicates with UEs 120 via a radio access link, such as an RU. In some examples, a network node 110 is or includes a network node that communicates with other network nodes 110 via a fronthaul link or a midhaul link, such as a DU. In some examples, a network node 110 is or includes a network node that communicates with other network nodes 110 via a midhaul link or a core network via a backhaul link, such as a CU. In some examples, a network node 110 (such as an aggregated network node 110 or a disaggregated network node 110) may include multiple network nodes, such as one or more RUs, one or more CUs, and/or one or more DUs. A network node 110 may include, for example, an NR base station, an LTE base station, a Node B, an eNB (e.g., in 4G), a gNB (e.g., in 5G), an access point, a transmission reception point (TRP), a DU, an RU, a CU, a mobility element of a network, a core network node, a network element, a network equipment, a RAN node, or a combination thereof. In some examples, the network nodes 110 may be interconnected to one another or to one or more other network nodes 110 in the wireless network 100 through various types of fronthaul, midhaul, and/or backhaul interfaces, such as a direct physical connection, an air interface, or a virtual network, using any suitable transport network.

[0039] In some examples, a network node 110 may provide communication coverage for a particular geographic area. In the Third Generation Partnership Project (3GPP), the term “cell” can refer to a coverage area of a network node 110 and/or a network node subsystem serving this coverage area, depending on the context in which the term

is used. A network node **110** may provide communication coverage for a macro cell, a pico cell, a femto cell, and/or another type of cell. A macro cell may cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs **120** with service subscriptions. A pico cell may cover a relatively small geographic area and may allow unrestricted access by UEs **120** with service subscriptions. A femto cell may cover a relatively small geographic area (e.g., a home) and may allow restricted access by UEs **120** having association with the femto cell (e.g., UEs **120** in a closed subscriber group (CSG)). A network node **110** for a macro cell may be referred to as a macro network node. A network node **110** for a pico cell may be referred to as a pico network node. A network node **110** for a femto cell may be referred to as a femto network node or an in-home network node. In the example shown in FIG. 1, the network node **110a** may be a macro network node for a macro cell **102a**, the network node **110b** may be a pico network node for a pico cell **102b**, and the network node **110c** may be a femto network node for a femto cell **102c**. A network node may support one or multiple (e.g., three) cells. In some examples, a cell may not necessarily be stationary, and the geographic area of the cell may move according to the location of a network node **110** that is mobile (e.g., a mobile network node).

[0040] In some aspects, the term “base station” or “network node” may refer to an aggregated base station, a disaggregated base station, an integrated access and backhaul (IAB) node, a relay node, or one or more components thereof. For example, in some aspects, “base station” or “network node” may refer to a CU, a DU, an RU, a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC, or a combination thereof. In some aspects, the term “base station” or “network node” may refer to one device configured to perform one or more functions, such as those described herein in connection with the network node **110**. In some aspects, the term “base station” or “network node” may refer to a plurality of devices configured to perform the one or more functions. For example, in some distributed systems, each of a quantity of different devices (which may be located in the same geographic location or in different geographic locations) may be configured to perform at least a portion of a function, or to duplicate performance of at least a portion of the function, and the term “base station” or “network node” may refer to any one or more of those different devices. In some aspects, the term “base station” or “network node” may refer to one or more virtual base stations or one or more virtual base station functions. For example, in some aspects, two or more base station functions may be instantiated on a single device. In some aspects, the term “base station” or “network node” may refer to one of the base station functions and not another. In this way, a single device may include more than one base station.

[0041] The wireless network **100** may include one or more relay stations. A relay station is a network node that can receive a transmission of data from an upstream node (e.g., a network node **110** or a UE **120**) and send a transmission of the data to a downstream node (e.g., a UE **120** or a network node **110**). A relay station may be a UE **120** that can relay transmissions for other UEs **120**. In the example shown in FIG. 1, the network node **110d** (e.g., a relay network node) may communicate with the network node **110a** (e.g., a macro network node) and the UE **120d** in order to facilitate

communication between the network node **110a** and the UE **120d**. A network node **110** that relays communications may be referred to as a relay station, a relay base station, a relay network node, a relay node, a relay, or the like.

[0042] The wireless network **100** may be a heterogeneous network that includes network nodes **110** of different types, such as macro network nodes, pico network nodes, femto network nodes, relay network nodes, or the like. These different types of network nodes **110** may have different transmit power levels, different coverage areas, and/or different impacts on interference in the wireless network **100**. For example, macro network nodes may have a high transmit power level (e.g., 5 to 40 watts) whereas pico network nodes, femto network nodes, and relay network nodes may have lower transmit power levels (e.g., 0.1 to 2 watts).

[0043] A network controller **130** may couple to or communicate with a set of network nodes **110** and may provide coordination and control for these network nodes **110**. The network controller **130** may communicate with the network nodes **110** via a backhaul communication link or a midhaul communication link. The network nodes **110** may communicate with one another directly or indirectly via a wireless or wireline backhaul communication link. In some aspects, the network controller **130** may be a CU or a core network device, or may include a CU or a core network device.

[0044] The UEs **120** may be dispersed throughout the wireless network **100**, and each UE **120** may be stationary or mobile. A UE **120** may include, for example, an access terminal, a terminal, a mobile station, and/or a subscriber unit. A UE **120** may be a cellular phone (e.g., a smart phone), a personal digital assistant (PDA), a wireless modem, a wireless communication device, a handheld device, a laptop computer, a cordless phone, a wireless local loop (WLL) station, a tablet, a camera, a gaming device, a netbook, a smartbook, an ultrabook, a medical device, a biometric device, a wearable device (e.g., a smart watch, smart clothing, smart glasses, a smart wristband, smart jewelry (e.g., a smart ring or a smart bracelet)), an entertainment device (e.g., a music device, a video device, and/or a satellite radio), a vehicular component or sensor, a smart meter/sensor, industrial manufacturing equipment, a global positioning system device, a UE function of a network node, and/or any other suitable device that is configured to communicate via a wireless or wired medium.

[0045] Some UEs **120** may be considered machine-type communication (MTC) or evolved or enhanced machine-type communication (eMTC) UEs. An MTC UE and/or an eMTC UE may include, for example, a robot, a drone, a remote device, a sensor, a meter, a monitor, and/or a location tag, that may communicate with a network node, another device (e.g., a remote device), or some other entity. Some UEs **120** may be considered Internet-of-Things (IOT) devices, and/or may be implemented as NB-IOT (narrow-band IoT) devices. Some UEs **120** may be considered a Customer Premises Equipment. A UE **120** may be included inside a housing that houses components of the UE **120**, such as processor components and/or memory components. In some examples, the processor components and the memory components may be coupled together. For example, the processor components (e.g., one or more processors) and the memory components (e.g., a memory) may be operatively coupled, communicatively coupled, electronically coupled, and/or electrically coupled.

[0046] In general, any number of wireless networks **100** may be deployed in a given geographic area. Each wireless network **100** may support a particular RAT and may operate on one or more frequencies. A RAT may be referred to as a radio technology, an air interface, or the like. A frequency may be referred to as a carrier, a frequency channel, or the like. Each frequency may support a single RAT in a given geographic area in order to avoid interference between wireless networks of different RATs. In some cases, NR or 5G RAT networks may be deployed.

[0047] In some examples, two or more UEs **120** (e.g., shown as UE **120a** and UE **120e**) may communicate directly using one or more sidelink channels (e.g., without using a network node **110** as an intermediary to communicate with one another). For example, the UEs **120** may communicate using peer-to-peer (P2P) communications, device-to-device (D2D) communications, a vehicle-to-everything (V2X) protocol (e.g., which may include a vehicle-to-vehicle (V2V) protocol, a vehicle-to-infrastructure (V2I) protocol, or a vehicle-to-pedestrian (V2P) protocol), and/or a mesh network. In such examples, a UE **120** may perform scheduling operations, resource selection operations, and/or other operations described elsewhere herein as being performed by the network node **110**.

[0048] Devices of the wireless network **100** may communicate using the electromagnetic spectrum, which may be subdivided by frequency or wavelength into various classes, bands, channels, or the like. For example, devices of the wireless network **100** may communicate using one or more operating bands. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

[0049] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0050] With the above examples in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like, if used herein, may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like, if used herein, may broadly represent frequencies that may include mid-

band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band. It is contemplated that the frequencies included in these operating bands (e.g., FR1, FR2, FR3, FR4, FR4-a, FR4-1, and/or FR5) may be modified, and techniques described herein are applicable to those modified frequency ranges.

[0051] In some aspects, the UE **120** may include a communication manager **140**. As described in more detail elsewhere herein, the communication manager **140** may receive an indication of an upcoming backscatter reading operation and adjust a reception parameter of the UE **120** in response to the indication. Additionally, or alternatively, the communication manager **140** may receive an indication of an upcoming backscatter reading operation and activate a low-power mode for one or more hardware components of the UE **120** in response to the indication. Additionally, or alternatively, the communication manager **140** may measure a set of beamformed reference signals associated with an upcoming backscatter reading operation and transmit a report indicating a set of measurements of the set of beamformed reference signals. Additionally, or alternatively, the communication manager **140** may perform one or more other operations described herein.

[0052] In some aspects, the network node **110** may include a communication manager **150**. As described in more detail elsewhere herein, the communication manager **150** may transmit an indication of an upcoming backscatter reading operation; and transmit a continuous wave (CW) signal to enable the backscatter reading operation. Additionally, or alternatively, the communication manager **150** may perform one or more other operations described herein.

[0053] As indicated above, FIG. 1 is provided as an example. Other examples may differ from what is described with regard to FIG. 1.

[0054] FIG. 2 is a diagram illustrating an example **200** of a network node **110** in communication with a UE **120** in a wireless network **100**, in accordance with the present disclosure. The network node **110** may be equipped with a set of antennas **234a** through **234t**, such as T antennas ($T \geq 1$). The UE **120** may be equipped with a set of antennas **252a** through **252r**, such as R antennas ($R \geq 1$). The network node **110** of example **200** includes one or more radio frequency components, such as antennas **234** and a modem **254**. In some examples, a network node **110** may include an interface, a communication component, or another component that facilitates communication with the UE **120** or another network node. Some network nodes **110** may not include radio frequency components that facilitate direct communication with the UE **120**, such as one or more CUs, or one or more DUs.

[0055] At the network node **110**, a transmit processor **220** may receive data, from a data source **212**, intended for the UE **120** (or a set of UEs **120**). The transmit processor **220** may select one or more modulation and coding schemes (MCSs) for the UE **120** based at least in part on one or more channel quality indicators (CQIs) received from that UE **120**. The network node **110** may process (e.g., encode and modulate) the data for the UE **120** based at least in part on the MCS(s) selected for the UE **120** and may provide data symbols for the UE **120**. The transmit processor **220** may process system information (e.g., for semi-static resource partitioning information (SRPI)) and control information (e.g., CQI requests, grants, and/or upper layer signaling) and provide overhead symbols and control symbols. The trans-

mit processor 220 may generate reference symbols for reference signals (e.g., a cell-specific reference signal (CRS) or a demodulation reference signal (DMRS)) and synchronization signals (e.g., a primary synchronization signal (PSS) or a secondary synchronization signal (SSS)). A transmit (TX) multiple-input multiple-output (MIMO) processor 230 may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, the overhead symbols, and/or the reference symbols, if applicable, and may provide a set of output symbol streams (e.g., T output symbol streams) to a corresponding set of modems 232 (e.g., T modems), shown as modems 232a through 232t. For example, each output symbol stream may be provided to a modulator component (shown as MOD) of a modem 232. Each modem 232 may use a respective modulator component to process a respective output symbol stream (e.g., for OFDM) to obtain an output sample stream. Each modem 232 may further use a respective modulator component to process (e.g., convert to analog, amplify, filter, and/or upconvert) the output sample stream to obtain a downlink signal. The modems 232a through 232t may transmit a set of downlink signals (e.g., T downlink signals) via a corresponding set of antennas 234 (e.g., T antennas), shown as antennas 234a through 234t.

[0056] At the UE 120, a set of antennas 252 (shown as antennas 252a through 252r) may receive the downlink signals from the network node 110 and/or other network nodes 110 and may provide a set of received signals (e.g., R received signals) to a set of modems 254 (e.g., R modems), shown as modems 254a through 254r. For example, each received signal may be provided to a demodulator component (shown as DEMOD) of a modem 254. Each modem 254 may use a respective demodulator component to condition (e.g., filter, amplify, downconvert, and/or digitize) a received signal to obtain input samples. Each modem 254 may use a demodulator component to further process the input samples (e.g., for OFDM) to obtain received symbols. A MIMO detector 256 may obtain received symbols from the modems 254, may perform MIMO detection on the received symbols if applicable, and may provide detected symbols. A receive processor 258 may process (e.g., demodulate and decode) the detected symbols, may provide decoded data for the UE 120 to a data sink 260, and may provide decoded control information and system information to a controller/processor 280. The term “controller/processor” may refer to one or more controllers, one or more processors, or a combination thereof. A channel processor may determine a reference signal received power (RSRP) parameter, a received signal strength indicator (RSSI) parameter, a reference signal received quality (RSRQ) parameter, and/or a CQI parameter, among other examples. In some examples, one or more components of the UE 120 may be included in a housing 284.

[0057] The network controller 130 may include a communication unit 294, a controller/processor 290, and a memory 292. The network controller 130 may include, for example, one or more devices in a core network. The network controller 130 may communicate with the network node 110 via the communication unit 294.

[0058] One or more antennas (e.g., antennas 234a through 234t and/or antennas 252a through 252r) may include, or may be included within, one or more antenna panels, one or more antenna groups, one or more sets of antenna elements, and/or one or more antenna arrays, among other examples.

An antenna panel, an antenna group, a set of antenna elements, and/or an antenna array may include one or more antenna elements (within a single housing or multiple housings), a set of coplanar antenna elements, a set of non-coplanar antenna elements, and/or one or more antenna elements coupled to one or more transmission and/or reception components, such as one or more components of FIG. 2.

[0059] On the uplink, at the UE 120, a transmit processor 264 may receive and process data from a data source 262 and control information (e.g., for reports that include RSRP, RSSI, RSRQ, and/or CQI) from the controller/processor 280. The transmit processor 264 may generate reference symbols for one or more reference signals. The symbols from the transmit processor 264 may be precoded by a TX MIMO processor 266 if applicable, further processed by the modems 254 (e.g., for DFT-s-OFDM or CP-OFDM), and transmitted to the network node 110. In some examples, the modem 254 of the UE 120 may include a modulator and a demodulator. In some examples, the UE 120 includes a transceiver. The transceiver may include any combination of the antenna(s) 252, the modem(s) 254, the MIMO detector 256, the receive processor 258, the transmit processor 264, and/or the TX MIMO processor 266. The transceiver may be used by a processor (e.g., the controller/processor 280) and the memory 282 to perform aspects of any of the methods described herein (e.g., with reference to FIGS. 6-14).

[0060] At the network node 110, the uplink signals from UE 120 and/or other UEs may be received by the antennas 234, processed by the modem 232 (e.g., a demodulator component, shown as DEMOD, of the modem 232), detected by a MIMO detector 236 if applicable, and further processed by a receive processor 238 to obtain decoded data and control information sent by the UE 120. The receive processor 238 may provide the decoded data to a data sink 239 and provide the decoded control information to the controller/processor 240. The network node 110 may include a communication unit 244 and may communicate with the network controller 130 via the communication unit 244. The network node 110 may include a scheduler 246 to schedule one or more UEs 120 for downlink and/or uplink communications. In some examples, the modem 232 of the network node 110 may include a modulator and a demodulator. In some examples, the network node 110 includes a transceiver. The transceiver may include any combination of the antenna(s) 234, the modem(s) 232, the MIMO detector 236, the receive processor 238, the transmit processor 220, and/or the TX MIMO processor 230. The transceiver may be used by a processor (e.g., the controller/processor 240) and the memory 242 to perform aspects of any of the methods described herein (e.g., with reference to FIGS. 6-14).

[0061] The controller/processor 240 of the network node 110, the controller/processor 280 of the UE 120, and/or any other component(s) of FIG. 2 may perform one or more techniques associated with indications and reporting for backscatter reading operations, as described in more detail elsewhere herein. For example, the controller/processor 240 of the network node 110, the controller/processor 280 of the UE 120, and/or any other component(s) of FIG. 2 may perform or direct operations of, for example, process 900 of FIG. 9, process 1000 of FIG. 10, process 1100 of FIG. 11, process 1200 of FIG. 12, and/or other processes as described herein. The memory 242 and the memory 282 may store data and program codes for the network node 110 and the UE

120, respectively. In some examples, the memory 242 and/or the memory 282 may include a non-transitory computer-readable medium storing one or more instructions (e.g., code and/or program code) for wireless communication. For example, the one or more instructions, when executed (e.g., directly, or after compiling, converting, and/or interpreting) by one or more processors of the network node 110 and/or the UE 120, may cause the one or more processors, the UE 120, and/or the network node 110 to perform or direct operations of, for example, process 900 of FIG. 9, process 1000 of FIG. 10, process 1100 of FIG. 11, process 1200 of FIG. 12, and/or other processes as described herein. In some examples, executing instructions may include running the instructions, converting the instructions, compiling the instructions, and/or interpreting the instructions, among other examples. In some aspects, the transmitting device described herein is the UE 120, is included in the UE 120, includes one or more components of the UE 120 shown in FIG. 2, is the network node 110, is included in the network node 110, or includes one or more components of the network node 110 shown in FIG. 2. In some aspects, the receiving device described herein is the UE 120, is included in the UE 120, or includes one or more components of the UE 120 shown in FIG. 2.

[0062] In some aspects, a receiving device (e.g., a UE 120) may include means for receiving an indication of an upcoming backscatter reading operation; and/or means for adjusting a reception parameter of the receiving device in response to the indication. Additionally, or alternatively, the receiving device may include means for receiving an indication of an upcoming backscatter reading operation; and/or means for activating a low-power mode for one or more hardware components of the receiving device in response to the indication. Additionally, or alternatively, the receiving device may include means for measuring a set of beam-formed reference signals associated with an upcoming backscatter reading operation; and/or means for transmitting a report indicating a set of measurements of the set of beam-formed reference signals. In some aspects, the means for the receiving device to perform operations described herein may include, for example, one or more of communication manager 140, antenna 252, modem 254, MIMO detector 256, receive processor 258, transmit processor 264, TX MIMO processor 266, controller/processor 280, or memory 282.

[0063] In some aspects, a transmitting device (e.g., a UE 120 and/or a network node 110) may include means for transmitting an indication of an upcoming backscatter reading operation; and/or means for transmitting a CW signal to enable the backscatter reading operation. In some aspects, the means for the transmitting device to perform operations described herein may include, for example, one or more of communication manager 150, transmit processor 220, TX MIMO processor 230, modem 232, antenna 234, MIMO detector 236, receive processor 238, controller/processor 240, memory 242, or scheduler 246. Alternatively, the means for the transmitting device to perform operations described herein may include, for example, one or more of communication manager 140, antenna 252, modem 254, MIMO detector 256, receive processor 258, transmit processor 264, TX MIMO processor 266, controller/processor 280, or memory 282.

[0064] While blocks in FIG. 2 are illustrated as distinct components, the functions described above with respect to the blocks may be implemented in a single hardware,

software, or combination component or in various combinations of components. For example, the functions described with respect to the transmit processor 264, the receive processor 258, and/or the TX MIMO processor 266 may be performed by or under the control of the controller/processor 280.

[0065] As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

[0066] Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a RAN node, a core network node, a network element, a base station, or a network equipment may be implemented in an aggregated or disaggregated architecture. For example, a base station (such as a Node B (NB), an evolved NB (eNB), an NR BS, a 5G NB, an access point (AP), a TRP, or a cell, among other examples), or one or more units (or one or more components) performing base station functionality, may be implemented as an aggregated base station (also known as a standalone base station or a monolithic base station) or a disaggregated base station. “Network entity” or “network node” may refer to a disaggregated base station, or to one or more units of a disaggregated base station (such as one or more CUs, one or more DUs, one or more RUs, or a combination thereof).

[0067] An aggregated base station (e.g., an aggregated network node) may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node (e.g., within a single device or unit). A disaggregated base station (e.g., a disaggregated network node) may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more CUs, one or more DUs, or one or more RUs). In some examples, a CU may be implemented within a network node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other network nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units, such as a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU), among other examples.

[0068] Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an IAB network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)) to facilitate scaling of communication systems by separating base station functionality into one or more units that can be individually deployed. A disaggregated base station may include functionality implemented across two or more units at various physical locations, as well as functionality implemented for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station can be configured for wired or wireless communication with at least one other unit of the disaggregated base station.

[0069] FIG. 3 is a diagram illustrating an example disaggregated base station architecture 300, in accordance with the present disclosure. The disaggregated base station archi-

structure 300 may include a CU 310 that can communicate directly with a core network 320 via a backhaul link, or indirectly with the core network 320 through one or more disaggregated control units (such as a Near-RT RIC 325 via an E2 link, or a Non-RT RIC 315 associated with a Service Management and Orchestration (SMO) Framework 305, or both). A CU 310 may communicate with one or more DUs 330 via respective midhaul links, such as through F1 interfaces. Each of the DUs 330 may communicate with one or more RUs 340 via respective fronthaul links. Each of the RUs 340 may communicate with one or more UEs 120 via respective radio frequency (RF) access links. In some implementations, a UE 120 may be simultaneously served by multiple RUs 340.

[0070] Each of the units, including the CUs 310, the DUs 330, the RUs 340, as well as the Near-RT RICs 325, the Non-RT RICs 315, and the SMO Framework 305, may include one or more interfaces or be coupled with one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to one or multiple communication interfaces of the respective unit, can be configured to communicate with one or more of the other units via the transmission medium. In some examples, each of the units can include a wired interface, configured to receive or transmit signals over a wired transmission medium to one or more of the other units, and a wireless interface, which may include a receiver, a transmitter or transceiver (such as an RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units. In some aspects, the CU 310 may host one or more higher layer control

[0071] functions. Such control functions can include radio resource control (RRC) functions, packet data convergence protocol (PDCP) functions, or service data adaptation protocol (SDAP) functions, among other examples. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU 310. The CU 310 may be configured to handle user plane functionality (for example, Central Unit-User Plane (CU-UP) functionality), control plane functionality (for example, Central Unit-Control Plane (CU-CP) functionality), or a combination thereof. In some implementations, the CU 310 can be logically split into one or more CU-UP units and one or more CU-CP units. A CU-UP unit can communicate bidirectionally with a CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU 310 can be implemented to communicate with a DU 330, as necessary, for network control and signaling.

[0072] Each DU 330 may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs 340. In some aspects, the DU 330 may host one or more of a radio link control (RLC) layer, a MAC layer, and one or more high physical (PHY) layers depending, at least in part, on a functional split, such as a functional split defined by the 3GPP. In some aspects, the one or more high PHY layers may be implemented by one or more modules for forward error correction (FEC) encoding and decoding, scrambling, and modulation and demodulation, among other examples. In some aspects, the DU 330 may further host one or more low PHY layers, such

as implemented by one or more modules for a fast Fourier transform (FFT), an inverse FFT (iFFT), digital beamforming, or physical random access channel (PRACH) extraction and filtering, among other examples. Each layer (which also may be referred to as a module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU 330, or with the control functions hosted by the CU 310.

[0073] Each RU 340 may implement lower-layer functionality. In some deployments, an RU 340, controlled by a DU 330, may correspond to a logical node that hosts RF processing functions or low-PHY layer functions, such as performing an FFT, performing an iFFT, digital beamforming, or PRACH extraction and filtering, among other examples, based on a functional split (for example, a functional split defined by the 3GPP), such as a lower layer functional split. In such an architecture, each RU 340 can be operated to handle over the air (OTA) communication with one or more UEs 120. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) 340 can be controlled by the corresponding DU 330. In some scenarios, this configuration can enable each DU 330 and the CU 310 to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0074] The SMO Framework 305 may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework 305 may be configured to support the deployment of dedicated physical resources for RAN coverage requirements, which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework 305 may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) platform 390) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs 310, DUs 330, RUs 340, non-RT RICs 315, and Near-RT RICs 325. In some implementations, the SMO Framework 305 can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) 311, via an O1 interface. Additionally, in some implementations, the SMO Framework 305 can communicate directly with each of one or more RUs 340 via a respective O1 interface. The SMO Framework 305 also may include a Non-RT RIC 315 configured to support functionality of the SMO Framework 305.

[0075] The Non-RT RIC 315 may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC 325. The Non-RT RIC 315 may be coupled to or communicate with (such as via an AI interface) the Near-RT RIC 325. The Near-RT RIC 325 may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs 310, one or more DUs 330, or both, as well as an O-eNB, with the Near-RT RIC 325.

[0076] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC 325, the Non-RT RIC 315 may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC 325 and may be received at the SMO Framework 305 or the Non-RT RIC 315 from non-network data sources or from network functions. In some examples, the Non-RT RIC 315 or the Near-RT RIC 325 may be configured to tune RAN behavior or performance. For example, the Non-RT RIC 315 may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework 305 (such as reconfiguration via an O1 interface) or via creation of RAN management policies (such as AI interface policies).

[0077] As indicated above, FIG. 3 is provided as an example. Other examples may differ from what is described with regard to FIG. 3.

[0078] FIG. 4 is a diagram illustrating an example 400 of a backscatter reading operation, in accordance with the present disclosure. Example 400 includes a transmitting (TX) device 401 (e.g., a UE 120 or a network node 110), a receiving (RX) device 403 (e.g., a UE 120), and a tag 405 (or another passive or semi-active device that backscatters signal from the TX device 401).

[0079] As shown in FIG. 4, the TX device 401 may communicate with the RX device 403 over a direct link. For example, the TX device 401 may communicate over a PC5 interface (e.g., associated with a sidelink channel) with the RX device 403 or over a Uu interface with the RX device 403. In example 400, a signal over the direct link is represented by hD1D2(n).

[0080] Additionally, or alternatively, as shown in FIG. 4, the TX device 401 may transmit a signal (e.g., a CW signal) to the tag 405. In example 400, the CW signal is represented by hD1T(n). The tag 405 may therefore backscatter the CW signal, resulting in a backscattered signal arriving at the RX device 403. In example 400, the signal transmitted by the tag 405 is represented by hTD2(n).

[0081] Accordingly, the RX device 403 may decode the backscattered signal to receive information from the tag 405. Alternatively, the RX device 403 may decode a superposition of the backscattered signal from the tag 405 and a signal over the direct link from the TX device 401.

[0082] FIG. 5 is a diagram illustrating an example 500 of the signals for a backscatter reading operation (e.g., the backscatter reading operation of FIG. 4), in accordance with the present disclosure. FIG. 5 shows an example graph of power for a CW signal 501 received at the RX device 403 (e.g., represented by hD1D2(n)) over time. FIG. 5 further shows an example graph of power for the backscattered signal 503 arriving at the RX device 403. In example 500, the signal arriving at the RX device 403 is a combination of the CW signal received at the tag 405 (e.g., represented by h1T(n)), the backscattered signal transmitted by the tag 405 (e.g., represented by hTD2(n)), an information signal applied by the tag 405 (e.g., represented by s(n)), and a power loss factor due to absorption of the CW signal at the tag 405 (e.g., represented by of). Accordingly, as shown in FIG. 5, the RX device 403 may decode the combined signal 505 that is received, which is a combination of the CW signal 501 received at the RX device 403 (e.g., represented by hD1D2(n)) and the backscattered signal arriving at the RX device 403.

[0083] In order to activate a tag, a TX device generally uses a CW signal with a much larger transmit power than other signals. However, decoding a combined signal is particularly difficult when the CW signal dominates receiving hardware at an RX device (e.g., an analog-to-digital converter (ADC), a low noise amplifier (LNA), an automatic gain control (AGC) loop, and/or components in an RF front end (RFFE), among other examples). Additionally, the CW signal may be strong enough to damage receiving hardware at the RX device and/or at other devices (e.g., other UEs) nearby the RX device.

[0084] Some techniques and apparatuses described herein enable a TX device (e.g., TX device 401) to indicate an upcoming backscatter reading operation. Accordingly, an RX device (e.g., RX device 403) may adjust a receiving parameter to improve accuracy when decoding the combined signal. As a result, the RX device 403 conserves power and processing resources that would otherwise have been wasted on a failed decoding attempt and on a subsequent retransmission by the TX device 401. Additionally, a nearby device may activate a low-power mode for one or more hardware components during the backscatter reading operation. As a result, the nearby device prevents possible hardware failure during the backscatter reading operation.

[0085] Additionally, or alternatively, some techniques and apparatuses described herein enable the TX device 401 to request measurements associated with different beamforming directions (also referred to as “beams”). Accordingly, the RX device 403 (and/or a nearby device) may report measurements of the beams. As a result, the TX device 401 may select a beam for a CW signal that minimizes chances of damage to hardware components of the RX device 403 and any nearby devices.

[0086] As indicated above, FIGS. 4 and 5 are provided as an example. Other examples may differ from what is described with respect to FIGS. 4 and 5.

[0087] FIG. 6 is a diagram illustrating an example 600 associated with filters for backscatter reading operations, in accordance with the present disclosure. In example 600, a TX device 401 (e.g., a UE 120 or a network node 110) may indicate an upcoming backscatter reading operation to an RX device 403 (e.g., a UE 120). The indication may be a layer 1 (L1) indication (e.g., downlink control information (DCI) or sidelink control information (SCI)), a layer 2 (L2) indication (e.g., a medium access control (MAC) control element (MAC-CE) or a sidelink MAC-CE), and/or a layer 3 (L3) indication (e.g., an RRC message or a sidelink RRC message). Although described in connection with the TX device 401 and the RX device 403 communicating over a Uu interface or a PC5 interface, the TX device 401 and the RX device 403 may alternatively communicate over a different interface that allows the TX device 401 to read a tag (e.g., tag 405) in coordination with the RX device 403.

[0088] Accordingly, the TX device 401 may transmit a CW signal that has a power 601, as shown in FIG. 6, in order to initiate the backscatter reading operation. Therefore, the RX device 403 may activate an analog stopband filter based on one or more CW frequencies. For example, the TX device 401 may indicate the one or more CW frequencies when indicating the upcoming backscatter reading operation.

[0089] The backscatter signal to be decoded by the RX device 403 may have a power 603a in a first set of frequencies and a power 603b in a second set of frequencies, as shown in FIG. 6. Additionally, or alternatively, the RX

device **403** may activate an analog passband filter based on one or more desired frequencies (e.g., the first set of frequencies and the second set of frequencies). For example, the TX device **401** may indicate the one or more desired frequencies when indicating the upcoming backscatter reading operation.

[0090] The RX device **403** may transmit a capability message to the TX device **401** that indicates settings associated with an analog passband of the RX device **403** and/or settings associated with an analog stopband of the RX device **403**. For example, the RX device **403** may indicate the bandwidths and/or the filtering values available for the analog stopband and/or the analog passband. Accordingly, the TX device **401** may indicate the bandwidths and/or the filtering values to use when indicating the upcoming backscatter reading operation.

[0091] By using techniques as described in connection with FIG. 6, the TX device **401** to indicate an upcoming backscatter reading operation. Accordingly, the RX device **403** adjusts a receiving parameter to improve accuracy when decoding the combined signal. As a result, the RX device **403** conserves power and processing resources that would otherwise have been wasted on a failed decoding attempt and on a subsequent retransmission by the TX device **401**.

[0092] As indicated above, FIG. 6 is provided as an example. Other examples may differ from what is described with respect to FIG. 6.

[0093] FIG. 7 is a diagram illustrating an example **700** associated with reporting for backscatter reading operations, in accordance with the present disclosure. In example **700**, a TX device **401** (e.g., a UE **120** or a network node **110**) may indicate an upcoming backscatter reading operation to an RX device **403** (e.g., a UE **120**). The indication may be an L1 indication, an L2 indication, and/or an L3 indication. Although described in connection with the TX device **401** and the RX device **403** communicating over a Uu interface or a PC5 interface, the TX device **401** and the RX device **403** may alternatively communicate over a different interface that allows the TX device **401** to read a tag (e.g., tag **405**) in coordination with the RX device **403**.

[0094] Accordingly, the RX device **403** (or a nearby device) may perform one or more measurements (e.g., on one or more reference signals from the TX device **401**) and transmit a report to the TX device **401** based on the measurement(s). For example, the report may indicate a power delta **701** between a CW signal (e.g., a reference signal, such as a channel state information reference signal (CSI-RS) or a sidelink CSI-RS (SL CSI-RS)) and a backscatter signal to be decoded (e.g., caused by the reference signal). Accordingly, the TX device **401** may reduce a transmit power associated with the CW signal when the RX device **403** indicates that the CW signal is overwhelming the hardware of the RX device **403** (e.g., based on the power delta **701** and/or another measurement value in the report). Conversely, the TX device **401** may increase a transmit power associated with the CW signal when the RX device **403** indicates that the backscatter signal cannot be detected or decoded (e.g., based on the power delta **701** and/or another measurement value in the report).

[0095] Although shown with respect to the backscatter signal, the power delta **701** may be measured with respect to other reference signals (or a direct signal from the TX device **401**). Similarly, although described with respect to a delta, the report may additionally or alternatively include a ratio

between a power of the CW signal and a power of the backscatter signal (or another reference signal or a direct signal from the TX device **401**) and/or a headroom value (e.g., based on any power delta described herein).

[0096] Additionally, or alternatively, the RX device **403** may transmit a power control command to the RX device **403**. For example, the RX device **403** may indicate to reduce a transmit power associated with the CW signal when the CW signal is overwhelming the hardware of the RX device **403**. Conversely, the RX device **403** may indicate to increase a transmit power associated with the CW signal when the backscatter signal cannot be detected or decoded.

[0097] In some aspects, the TX device **401** may additionally or alternatively transmit a configuration associated with a set of beamformed reference signals associated with the upcoming backscatter reading operation. Accordingly, the RX device **403** (and/or a nearby device) may perform measurements on the set of beamformed reference signals and report RSRP (or sidelink RSRP) values, RSRQ (or sidelink RSRQ) values, and/or signal-to-interference-plus-noise ratio (SINR) (or sidelink SINR) values to the TX device **401**. Accordingly, the TX device **401** may select a beam associated with a lowest measurement value (e.g., at the RX device **403** and/or at a nearby device) in order to reduce chances of damage to hardware components of the RX device **403** and any nearby devices.

[0098] By using techniques as described in connection with FIG. 7, the TX device **401** requests measurements associated with the backscatter reading operation. As a result, the TX device **401** may apply power control for the CW signal that minimizes chances of damage to hardware components of the RX device **403** and any nearby devices.

[0099] As indicated above, FIG. 7 is provided as an example. Other examples may differ from what is described with respect to FIG. 7.

[0100] FIG. 8 is a diagram illustrating an example **800** associated with performing a backscatter reading operation, in accordance with the present disclosure. As shown in FIG. 8, a TX device **401** may perform a backscatter reading operation using a tag **405** and an RX device **403**. A nearby device **801** (e.g., a UE **120** or a network node **110**) may be near the tag **405** and/or the RX device **403**. Although described in connection with the TX device **401** and the RX device **403** communicating over a Uu interface or a PC5 interface, the TX device **401** and the RX device **403** may alternatively communicate over a different interface that allows the TX device **401** to read the tag **405** in coordination with the RX device **403**.

[0101] In some aspects, the TX device **401** may select the RX device **403** to perform the backscatter reading operation based on a location of the RX device **403** relative to one or more locations of one or more devices near the receiving device, such as the nearby device **801**. Accordingly, the TX device **401** may select an RX device **403** that is furthest from a set of nearby devices (e.g., based on median distance, a clustering factor, and/or another similar measure) in order to reduce chances of damage to as many of the set of nearby device as possible.

[0102] As shown by reference number **805**, the TX device **401** may transmit an indication of an upcoming backscatter reading operation. For example, the indication may be an L1 indication, an L2 indication, and/or an L3 indication. The RX device **403** may receive the indication along with the nearby device **801**.

[0103] Accordingly, as shown by reference number 810, the RX device 403 may adjust a reception parameter in response to the indication. For example, the RX device 403 may enable an analog stopband filter and/or an analog passband filter, as described in connection with FIG. 6. Additionally, or alternatively, the RX device 403 may change an ADC to operate in a different (e.g., higher) dynamic range in response to the indication.

[0104] Further, as shown by reference number 815, the nearby device 801 may activate a low-power mode for one or more hardware components of the nearby device 801 in response to the indication. As used herein, a “low-power mode” refers to a mode in which a hardware component is not supplied power or is supplied power below a usual operating power. The nearby device 801 may activate the low-power mode for an LNA, an AGC loop, an RFFE component, and/or another hardware component associated with receiving and processing wireless signals. Additionally, in some aspects, the nearby device 801 may enable an analog stopband filter and/or an analog passband filter, as described in connection with FIG. 6.

[0105] Accordingly, as shown by reference number 820, the TX device 401 may transmit a CW signal to enable the backscatter reading operation. For example, the CW signal may be beamformed based on a report from the RX device 403 and/or the nearby device 801, as described in connection with FIG. 7. Additionally, the CW signal may have a transmit power determined based on a report from the RX device 403 and/or the nearby device 801, as described in connection with FIG. 7.

[0106] In some aspects, data programmed into the tag 405 may be associated with a priority and/or a quality-of-service (QoS). Accordingly, the TX device 401 may schedule the CW signal based on the priority and/or the QoS associated with the tag 405. Therefore, data that the TX device 401 is to transmit to the RX device 403 (and/or the nearby device 801) may be transmitted before or after the CW signal, depending on the priorities and/or the QoSs associated with the data to be transmitted and the tag 405.

[0107] As shown by reference number 825, the RX device 403 may receive a backscattered signal (e.g., from the tag 405) using the adjusted reception parameter. Accordingly, the RX device 403 may decode the data programmed into the tag 405 (and included in the backscattered signal, as described in connection with FIGS. 4 and 5). As shown by reference number 830, the RX device 403 may relay the decoded data from the tag 405 to the TX device 401.

[0108] By using techniques as described in connection with FIG. 8, the TX device 401 indicates an upcoming backscatter reading operation. Accordingly, the RX device 403 may adjust a receiving parameter to improve accuracy when decoding the combined signal. As a result, the RX device 403 conserves power and processing resources that would otherwise have been wasted on a failed decoding attempt and on a subsequent retransmission by the TX device 401. Additionally, the nearby device 801 may activate a low-power mode for one or more hardware components during the backscatter reading operation. As a result, the nearby device 801 prevents possible hardware failure during the backscatter reading operation.

[0109] As indicated above, FIG. 8 is provided as an example. Other examples may differ from what is described with respect to FIG. 8.

[0110] FIG. 9 is a diagram illustrating an example process 900 performed, for example, by a receiving device, in accordance with the present disclosure. Example process 900 is an example where the receiving device (e.g., RX device 403 and/or apparatus 1300 of FIG. 13) performs operations associated with indications for backscatter reading operations.

[0111] As shown in FIG. 9, in some aspects, process 900 may include receiving an indication of an upcoming backscatter reading operation (block 910). For example, the receiving device (e.g., using communication manager 140 and/or reception component 1302, depicted in FIG. 13) may receive an indication of an upcoming backscatter reading operation, as described herein.

[0112] As further shown in FIG. 9, in some aspects, process 900 may include adjusting a reception parameter of the receiving device in response to the indication (block 920). For example, the receiving device (e.g., using communication manager 140 and/or adjustment component 1308, depicted in FIG. 13) may adjust a reception parameter of the receiving device in response to the indication, as described herein.

[0113] Process 900 may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0114] In a first aspect, the indication is included in DCI, a MAC-CE, or an RRC message.

[0115] In a second aspect, alone or in combination with the first aspect, the indication is included in SCI, a sidelink MAC-CE, or a sidelink RRC message.

[0116] In a third aspect, alone or in combination with one or more of the first and second aspects, adjusting the reception parameter includes enabling an analog passband filter based on one or more desired frequencies, enabling an analog stopband filter based on one or more continuous wave frequencies, or a combination thereof.

[0117] In a fourth aspect, alone or in combination with one or more of the first through third aspects, process 900 includes transmitting (e.g., using communication manager 140 and/or transmission component 1304, depicted in FIG. 13) a capability message indicating settings associated with an analog passband of the receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, such that the indication of the upcoming backscatter reading operation is based on the capability message.

[0118] In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, adjusting the reception parameter includes adjusting a dynamic range associated with an ADC of the receiving device.

[0119] In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, process 900 includes transmitting (e.g., using communication manager 140 and/or transmission component 1304) a capability message indicating a dynamic range associated with an ADC of the receiving device, such that the indication of the upcoming backscatter reading operation is based on the capability message.

[0120] In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, process 900 includes receiving (e.g., using communication manager 140 and/or reception component 1302) a backscattered signal using the adjusted reception parameter, such that the back-

scattered signal is received based on at least one of an associated priority or an associated QoS.

[0121] Although FIG. 9 shows example blocks of process 900, in some aspects, process 900 may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 9. Additionally, or alternatively, two or more of the blocks of process 900 may be performed in parallel.

[0122] FIG. 10 is a diagram illustrating an example process 1000 performed, for example, by a receiving device, in accordance with the present disclosure. Example process 1000 is an example where the receiving device (e.g., RX device 403 and/or apparatus 1300 of FIG. 13) performs operations associated with indications for backscatter reading operations.

[0123] As shown in FIG. 10, in some aspects, process 1000 may include receiving an indication of an upcoming backscatter reading operation (block 1010). For example, the receiving device (e.g., using communication manager 140 and/or reception component 1302, depicted in FIG. 13) may receive an indication of an upcoming backscatter reading operation, as described herein.

[0124] As further shown in FIG. 10, in some aspects, process 1000 may include

[0125] activating a low-power mode for one or more hardware components of the receiving device in response to the indication (block 1020). For example, the receiving device (e.g., using communication manager 140 and/or power management component 1310, depicted in FIG. 13) may activate a low-power mode for one or more hardware components of the receiving device in response to the indication, as described herein.

[0126] Process 1000 may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0127] In a first aspect, the indication is included in DCI, a MAC-CE, or an RRC message.

[0128] In a second aspect, alone or in combination with the first aspect, the indication is included in SCI, a sidelink MAC-CE, or a sidelink RRC message.

[0129] In a third aspect, alone or in combination with one or more of the first and second aspects, the indication identifies one or more time resources, one or more frequency resources, or a combination thereof.

[0130] In a fourth aspect, alone or in combination with one or more of the first through third aspects, process 1000 includes enabling (e.g., using communication manager 140 and/or adjustment component 1308, depicted in FIG. 13) an analog passband filter based on one or more desired frequencies, enabling (e.g., using communication manager 140 and/or adjustment component 1308) an analog stopband filter based on one or more CW frequencies, or a combination thereof.

[0131] Although FIG. 10 shows example blocks of process 1000, in some aspects, process 1000 may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 10. Additionally, or alternatively, two or more of the blocks of process 1000 may be performed in parallel.

[0132] FIG. 11 is a diagram illustrating an example process 1100 performed, for example, by a receiving device, in accordance with the present disclosure. Example process 1100 is an example where the receiving device (e.g., RX

device 403 and/or apparatus 1300 of FIG. 13) performs operations associated with reporting for backscatter reading operations.

[0133] As shown in FIG. 11, in some aspects, process 1100 may include measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation (block 1110). For example, the receiving device (e.g., using communication manager 140 and/or measurement component 1312, depicted in FIG. 13) may measure a set of beamformed reference signals associated with an upcoming backscatter reading operation, as described herein.

[0134] As further shown in FIG. 11, in some aspects, process 1100 may include transmitting a report indicating a set of measurements of the set of beamformed reference signals (block 1120). For example, the receiving device (e.g., using communication manager 140 and/or transmission component 1304, depicted in FIG. 13) may transmit a report indicating a set of measurements of the set of beamformed reference signals, as described herein.

[0135] Process 1100 may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0136] In a first aspect, the set of measurements includes one or more RSRP values, one or more RSRQ values, one or more SINR values, or a combination thereof.

[0137] In a second aspect, alone or in combination with the first aspect, the report indicates a ratio between a direct signal and a continuous wave signal.

[0138] In a third aspect, alone or in combination with one or more of the first and second aspects, the report indicates a delta associated with a CW signal with respect to a reference signal.

[0139] In a fourth aspect, alone or in combination with one or more of the first through third aspects, the report indicates a delta associated with a CW signal with respect to a direct signal.

[0140] In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, the report indicates a headroom associated with a CW signal.

[0141] In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, the report comprises a power control command.

[0142] Although FIG. 11 shows example blocks of process 1100, in some aspects, process 1100 may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 11. Additionally, or alternatively, two or more of the blocks of process 1100 may be performed in parallel.

[0143] FIG. 12 is a diagram illustrating an example process 1200 performed, for example, by a transmitting device, in accordance with the present disclosure. Example process 1200 is an example where the transmitting device (e.g., TX device 401 and/or apparatus 1400 of FIG. 14) performs operations associated with indications and reporting for backscatter reading operations.

[0144] As shown in FIG. 12, in some aspects, process 1200 may include transmitting an indication of an upcoming backscatter reading operation (block 1210). For example, the transmitting device (e.g., using communication manager 150 and/or transmission component 1404, depicted in FIG. 14) may transmit an indication of an upcoming backscatter reading operation, as described herein.

[0145] As further shown in FIG. 12, in some aspects, process 1200 may include transmitting a CW signal to enable the backscatter reading operation (block 1220). For example, the transmitting device (e.g., using communication manager 150 and/or transmission component 1404) may transmit a CW signal to enable the backscatter reading operation, as described herein.

[0146] Process 1200 may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0147] In a first aspect, the indication is transmitted to a receiving device that will perform the backscatter reading operation.

[0148] In a second aspect, alone or in combination with the first aspect, the receiving device is selected based on a location of the receiving device relative to one or more locations of one or more devices near the receiving device.

[0149] In a third aspect, alone or in combination with one or more of the first and second aspects, the indication is transmitted to one or more devices near a receiving device that will perform the backscatter reading operation.

[0150] In a fourth aspect, alone or in combination with one or more of the first through third aspects, process 1200 includes receiving (e.g., using communication manager 150 and/or reception component 1402, depicted in FIG. 14) a capability message indicating settings associated with an analog passband of a receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, such that the indication of the upcoming backscatter reading operation is based on the capability message.

[0151] In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, process 1200 includes receiving (e.g., using communication manager 150 and/or reception component 1402) a capability message indicating a dynamic range associated with an ADC of a receiving device, such that the indication of the upcoming backscatter reading operation is based on the capability message.

[0152] In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, process 1200 includes transmitting (e.g., using communication manager 150 and/or transmission component 1404) a set of beamformed reference signals associated with the upcoming backscatter reading operation, and receiving (e.g., using communication manager 150 and/or reception component 1402) a report indicating a set of measurements of the set of beamformed reference signals, such that the CW signal is transmitted based on the report.

[0153] In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, the report comprises a power control command.

[0154] Although FIG. 12 shows example blocks of process 1200, in some aspects, process 1200 may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 12. Additionally, or alternatively, two or more of the blocks of process 1200 may be performed in parallel.

[0155] FIG. 13 is a diagram of an example apparatus 1300 for wireless communication. The apparatus 1300 may be a receiving device, or a receiving device may include the apparatus 1300. In some aspects, the apparatus 1300 includes a reception component 1302 and a transmission

component 1304, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus 1300 may communicate with another apparatus 1306 (such as a UE, a base station, or another wireless communication device) using the reception component 1302 and the transmission component 1304. As further shown, the apparatus 1300 may include the communication manager 140. The communication manager 140 may include one or more of an adjustment component 1308, a power management component 1310, or a measurement component 1312, among other examples.

[0156] In some aspects, the apparatus 1300 may be configured to perform one or more operations described herein in connection with FIGS. 6-8. Additionally, or alternatively, the apparatus 1300 may be configured to perform one or more processes described herein, such as process 900 of FIG. 9, process 1000 of FIG. 10, process 1100 of FIG. 11, or a combination thereof. In some aspects, the apparatus 1300 and/or one or more components shown in FIG. 13 may include one or more components of the UE described in connection with FIG. 2. Additionally, or alternatively, one or more components shown in FIG. 13 may be implemented within one or more components described in connection with FIG. 2. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

[0157] The reception component 1302 may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus 1306. The reception component 1302 may provide received communications to one or more other components of the apparatus 1300. In some aspects, the reception component 1302 may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus 1300. In some aspects, the reception component 1302 may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2.

[0158] The transmission component 1304 may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus 1306. In some aspects, one or more other components of the apparatus 1300 may generate communications and may provide the generated communications to the transmission component 1304 for transmission to the apparatus 1306. In some aspects, the transmission component 1304 may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus 1306. In some aspects, the transmission component 1304 may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/

processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. In some aspects, the transmission component 1304 may be co-located with the reception component 1302 in a transceiver.

[0159] In some aspects, the reception component 1302 may receive an indication of an upcoming backscatter reading operation. Accordingly, the adjustment component 1308 may adjust a reception parameter of the apparatus 1300 in response to the indication. In some aspects, the adjustment component 1308 may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. In some aspects, the transmission component 1304 may transmit a capability message indicating settings associated with an analog passband of the apparatus 1300, settings associated with an analog stopband of the apparatus 1300, or a combination thereof. Additionally, or alternatively, the transmission component 1304 may transmit a capability message indicating a dynamic range associated with an ADC of the apparatus 1300. The reception component 1302 may receive a backscattered signal, using the adjusted reception parameter, based on at least one of an associated priority or an associated QoS.

[0160] Additionally, or alternatively, the reception component 1302 may receive an indication of an upcoming backscatter reading operation. Accordingly, the power management component 1310 may activate a low-power mode for one or more hardware components of the apparatus 1300 in response to the indication. In some aspects, the power management component 1310 may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. In some aspects, the adjustment component 1308 may enable an analog passband filter based on one or more desired frequencies and/or enable an analog stopband filter based on one or more continuous wave frequencies.

[0161] Additionally, or alternatively, the measurement component 1312 may measure a set of beamformed reference signals associated with an upcoming backscatter reading operation. In some aspects, the measurement component 1312 may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. Accordingly, the transmission component 1304 may transmit a report indicating a set of measurements of the set of beamformed reference signals.

[0162] The number and arrangement of components shown in FIG. 13 are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. 13. Furthermore, two or more components shown in FIG. 13 may be implemented within a single component, or a single component shown in FIG. 13 may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. 13 may perform one or more functions described as being performed by another set of components shown in FIG. 13.

[0163] FIG. 14 is a diagram of an example apparatus 1400 for wireless communication. The apparatus 1400 may be a transmitting device, or a transmitting device may include the

apparatus 1400. In some aspects, the apparatus 1400 includes a reception component 1402 and a transmission component 1404, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus 1400 may communicate with another apparatus 1406 (such as a UE, a base station, or another wireless communication device) using the reception component 1402 and the transmission component 1404. As further shown, the apparatus 1400 may include the communication manager 150. The communication manager 150 may include a beamforming component 1408, among other examples.

[0164] In some aspects, the apparatus 1400 may be configured to perform one or more operations described herein in connection with FIGS. 6-8. Additionally, or alternatively, the apparatus 1400 may be configured to perform one or more processes described herein, such as process 1200 of FIG. 12, or a combination thereof. In some aspects, the apparatus 1400 and/or one or more components shown in FIG. 14 may include one or more components of the network node or of the UE described in connection with FIG. 2. Additionally, or alternatively, one or more components shown in FIG. 14 may be implemented within one or more components described in connection with FIG. 2. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

[0165] The reception component 1402 may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus 1406. The reception component 1402 may provide received communications to one or more other components of the apparatus 1400. In some aspects, the reception component 1402 may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus 1400. In some aspects, the reception component 1402 may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the network node or of the UE described in connection with FIG. 2.

[0166] The transmission component 1404 may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus 1406. In some aspects, one or more other components of the apparatus 1400 may generate communications and may provide the generated communications to the transmission component 1404 for transmission to the apparatus 1406. In some aspects, the transmission component 1404 may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus 1406. In some aspects, the transmission component 1404 may

include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the network node or of the UE described in connection with FIG. 2. In some aspects, the transmission component 1404 may be co-located with the reception component 1402 in a transceiver.

[0167] In some aspects, the transmission component 1404 may transmit an indication of an upcoming backscatter reading operation. Additionally, the transmission component 1404 may transmit a CW signal to enable the backscatter reading operation.

[0168] In some aspects, the reception component 1402 may receive a capability message indicating settings associated with an analog passband of a receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof. Additionally, or alternatively, the reception component 1402 may receive a capability message indicating a dynamic range associated with an analog-to-digital converter of a receiving device.

[0169] In some aspects, the transmission component 1404 may transmit a set of beamformed reference signals associated with the upcoming backscatter reading operation. For example, the beamforming component 1408 may generate control signals to form the set of beamformed reference signals. In some aspects, the beamforming component 1408 may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the network node or of the UE described in connection with FIG. 2. Accordingly, the reception component 1402 may receive a report indicating a set of measurements of the set of beamformed reference signals.

[0170] The number and arrangement of components shown in FIG. 14 are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. 14. Furthermore, two or more components shown in FIG. 14 may be implemented within a single component, or a single component shown in FIG. 14 may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. 14 may perform one or more functions described as being performed by another set of components shown in FIG. 14.

[0171] The following provides an overview of some Aspects of the present disclosure:

[0172] Aspect 1: A method of wireless communication performed by a receiving device, comprising: receiving an indication of an upcoming backscatter reading operation; and adjusting a reception parameter of the receiving device in response to the indication.

[0173] Aspect 2: The method of Aspect 1, wherein the indication is included in downlink control information, a medium access control layer control element, or a radio resource control message.

[0174] Aspect 3: The method of Aspect 1, wherein the indication is included in sidelink control information, a sidelink medium access control layer control element, or a sidelink radio resource control message.

[0175] Aspect 4: The method of any of Aspects 1 through 3, wherein adjusting the reception parameter comprises: enabling an analog passband filter based on one or more

desired frequencies; enabling an analog stopband filter based on one or more continuous wave frequencies; or a combination thereof.

[0176] Aspect 5: The method of any of Aspects 1 through 4, further comprising: transmitting a capability message indicating settings associated with an analog passband of the receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

[0177] Aspect 6: The method of any of Aspects 1 through 5, wherein adjusting the reception parameter comprises: adjusting a dynamic range associated with an analog-to-digital converter of the receiving device.

[0178] Aspect 7: The method of any of Aspects 1 through 6, further comprising: transmitting a capability message indicating a dynamic range associated with an analog-to-digital converter of the receiving device, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

[0179] Aspect 8: The method of any of Aspects 1 through 7, further comprising: receiving a backscattered signal using the adjusted reception parameter, wherein the backscattered signal is received based on at least one of an associated priority or an associated quality-of-service.

[0180] Aspect 9: A method of wireless communication performed by a receiving device, comprising: receiving an indication of an upcoming backscatter reading operation; and activating a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0181] Aspect 10: The method of Aspect 9, wherein the indication is included in downlink control information, a medium access control layer control element, or a radio resource control message.

[0182] Aspect 11: The method of Aspect 9, wherein the indication is included in sidelink control information, a sidelink medium access control layer control element, or a sidelink radio resource control message.

[0183] Aspect 12: The method of any of Aspects 9 through 11, wherein the indication identifies one or more time resources, one or more frequency resources, or a combination thereof.

[0184] Aspect 13: The method of any of Aspects 9 through 12, further comprising: enabling an analog passband filter based on one or more desired frequencies; enabling an analog stopband filter based on one or more continuous wave frequencies; or a combination thereof.

[0185] Aspect 14: A method of wireless communication performed by a receiving device, comprising: measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation; and transmitting a report indicating a set of measurements of the set of beamformed reference signals.

[0186] Aspect 15: The method of Aspect 14, wherein the set of measurements includes one or more reference signal received power values, one or more reference signal received quality values, one or more signal-to-interference-plus-noise ratio values, or a combination thereof.

[0187] Aspect 16: The method of any of Aspects 14 through 15, wherein the report indicates a ratio between a direct signal and a continuous wave signal.

[0188] Aspect 17: The method of any of Aspects 14 through 16, wherein the report indicates a delta associated with a continuous wave signal with respect to a reference signal.

[0189] Aspect 18: The method of any of Aspects 14 through 17, wherein the report indicates a delta associated with a continuous wave signal with respect to a direct signal.

[0190] Aspect 19: The method of any of Aspects 14 through 18, wherein the report indicates a headroom associated with a continuous wave signal.

[0191] Aspect 20: The method of any of Aspects 14 through 19, wherein the report comprises a power control command.

[0192] Aspect 21: A method of wireless communication performed by a transmitting device, comprising: transmitting an indication of an upcoming backscatter reading operation; and transmitting a continuous wave (CW) signal to enable the backscatter reading operation.

[0193] Aspect 22: The method of Aspect 21, wherein the indication is transmitted to a receiving device that will perform the backscatter reading operation.

[0194] Aspect 23: The method of Aspect 22, wherein the receiving device is selected based on a location of the receiving device relative to one or more locations of one or more devices near the receiving device.

[0195] Aspect 24: The method of any of Aspects 21 through 23, wherein the indication is transmitted to one or more devices near a receiving device that will perform the backscatter reading operation.

[0196] Aspect 25: The method of any of Aspects 21 through 24, further comprising: receiving a capability message indicating settings associated with an analog passband of a receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

[0197] Aspect 26: The method of any of Aspects 21 through 25, further comprising: receiving a capability message indicating a dynamic range associated with an analog-to-digital converter of a receiving device, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

[0198] Aspect 27: The method of any of Aspects 21 through 26, further comprising: transmitting a set of beamformed reference signals associated with the upcoming backscatter reading operation; and receiving a report indicating a set of measurements of the set of beamformed reference signals, wherein the CW signal is transmitted based on the report.

[0199] Aspect 28: The method of Aspect 27, wherein the report comprises a power control command.

[0200] Aspect 29: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 1-8.

[0201] Aspect 30: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 1-8.

[0202] Aspect 31: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 1-8.

[0203] Aspect 32: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 1-8.

[0204] Aspect 33: A non-transitory computer-readable medium storing a set of

[0205] instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 1-8.

[0206] Aspect 34: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 9-13.

[0207] Aspect 35: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 9-13.

[0208] Aspect 36: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 9-13.

[0209] Aspect 37: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 9-13.

[0210] Aspect 38: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 9-13.

[0211] Aspect 39: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 14-20.

[0212] Aspect 40: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 14-20.

[0213] Aspect 41: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 14-20.

[0214] Aspect 42: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 14-20.

[0215] Aspect 43: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 14-20.

[0216] Aspect 44: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 21-28.

[0217] Aspect 45: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 21-28.

[0218] Aspect 46: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 21-28.

[0219] Aspect 47: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 21-28.

[0220] Aspect 48: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 21-28.

[0221] The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the aspects to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the aspects.

[0222] As used herein, the term “component” is intended to be broadly construed as hardware and/or a combination of hardware and software. “Software” shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, and/or functions, among other examples, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. As used herein, a “processor” is implemented in hardware and/or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware and/or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the aspects. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code, since those skilled in the art will understand that software and hardware can be designed to implement the systems and/or methods based, at least in part, on the description herein.

[0223] As used herein, “satisfying a threshold” may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

[0224] Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various aspects. Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. The disclosure of various aspects includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).

[0225] No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and

“an” are intended to include one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the terms “set” and “group” are intended to include one or more items and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

What is claimed is:

1. An apparatus for wireless communication at a receiving device, comprising:

a memory; and

one or more processors, coupled to the memory, configured to:

receive an indication of an upcoming backscatter reading operation; and

adjust a reception parameter of the receiving device in response to the indication.

2. The apparatus of claim 1, wherein the indication is included in downlink control information, a medium access control layer control element, or a radio resource control message.

3. The apparatus of claim 1, wherein the indication is included in sidelink control information, a sidelink medium access control layer control element, or a sidelink radio resource control message.

4. The apparatus of claim 1, wherein, to adjust the reception parameter, the one or more processors are configured to:

enable an analog passband filter based on one or more desired frequencies;

enable an analog stopband filter based on one or more continuous wave frequencies; or

a combination thereof.

5. The apparatus of claim 1, wherein the one or more processors are further configured to:

transmit a capability message indicating settings associated with an analog passband of the receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof,

wherein the indication of the upcoming backscatter reading operation is based on the capability message.

6. The apparatus of claim 1, wherein, to adjust the reception parameter, the one or more processors are configured to:

adjust a dynamic range associated with an analog-to-digital converter of the receiving device.

7. The apparatus of claim 1, wherein the one or more processors are further configured to:

transmit a capability message indicating a dynamic range associated with an analog-to-digital converter of the receiving device,

wherein the indication of the upcoming backscatter reading operation is based on the capability message.

8. The apparatus of claim 1, wherein the one or more processors are further configured to:

receive a backscattered signal using the adjusted reception parameter,

wherein the backscattered signal is received based on at least one of an associated priority or an associated quality-of-service.

9. An apparatus for wireless communication at a receiving device, comprising:

a memory; and

one or more processors, coupled to the memory, configured to:

receive an indication of an upcoming backscatter reading operation; and

activate a low-power mode for one or more hardware components of the receiving device in response to the indication.

10. The apparatus of claim 9, wherein the indication is included in downlink control information, a medium access control layer control element, or a radio resource control message.

11. The apparatus of claim 9, wherein the indication is included in sidelink control information, a sidelink medium access control layer control element, or a sidelink radio resource control message.

12. The apparatus of claim 9, wherein the indication identifies one or more time resources, one or more frequency resources, or a combination thereof.

13. The apparatus of claim 9, wherein the one or more processors are further configured to:

enable an analog passband filter based on one or more desired frequencies;

enable an analog stopband filter based on one or more continuous wave frequencies; or

a combination thereof.

14. An apparatus for wireless communication at a receiving device, comprising:

a memory; and

one or more processors, coupled to the memory, configured to:

measure a set of beamformed reference signals associated with an upcoming backscatter reading operation; and

transmit a report indicating a set of measurements of the set of beamformed reference signals.

15. The apparatus of claim 14, wherein the set of measurements includes one or more reference signal received power values, one or more reference signal received quality values, one or more signal-to-interference-plus-noise ratio values, or a combination thereof.

16. The apparatus of claim 14, wherein the report indicates a ratio between a direct signal and a continuous wave signal.

17. The apparatus of claim 14, wherein the report indicates a delta associated with a continuous wave signal with respect to a reference signal.

18. The apparatus of claim 14, wherein the report indicates a delta associated with a continuous wave signal with respect to a direct signal.

19. The apparatus of claim 14, wherein the report indicates a headroom associated with a continuous wave signal.

20. The apparatus of claim 14, wherein the report comprises a power control command.

21. An apparatus for wireless communication at a transmitting device, comprising:

a memory; and

one or more processors, coupled to the memory, configured to:

transmit an indication of an upcoming backscatter reading operation; and

transmit a continuous wave (CW) signal to enable the backscatter reading operation.

22. The apparatus of claim 21, wherein the indication is transmitted to a receiving device that will perform the backscatter reading operation.

23. The apparatus of claim 22, wherein the receiving device is selected based on a location of the receiving device relative to one or more locations of one or more devices near the receiving device.

24. The apparatus of claim 21, wherein the indication is transmitted to one or more devices near a receiving device that will perform the backscatter reading operation.

25. The apparatus of claim 21, wherein the one or more processors are further configured to:

receive a capability message indicating settings associated with an analog passband of a receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof,

wherein the indication of the upcoming backscatter reading operation is based on the capability message.

26. The apparatus of claim 21, wherein the one or more processors are further configured to:

receive a capability message indicating a dynamic range associated with an analog-to-digital converter of a receiving device,

wherein the indication of the upcoming backscatter reading operation is based on the capability message.

27. The apparatus of claim 21, wherein the one or more processors are further configured to:

transmit a set of beamformed reference signals associated with the upcoming backscatter reading operation; and receive a report indicating a set of measurements of the set of beamformed reference signals,

wherein the CW signal is transmitted based on the report.

28. The apparatus of claim 27, wherein the report comprises a power control command.

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